UNIVERSITY OF LEEDS

SCHOOL OF CHEMICAL AND PROCESS ENGINEERING

CAPE3000 DESIGN PROJECT

PLANT EQUIPMENT DESIGN REPORT

POST ADDITION MIXER AND SPRAY NOZZLES

Group Number:

CE1

Programme of Study:

Chemical Engineering

Project Supervisor:

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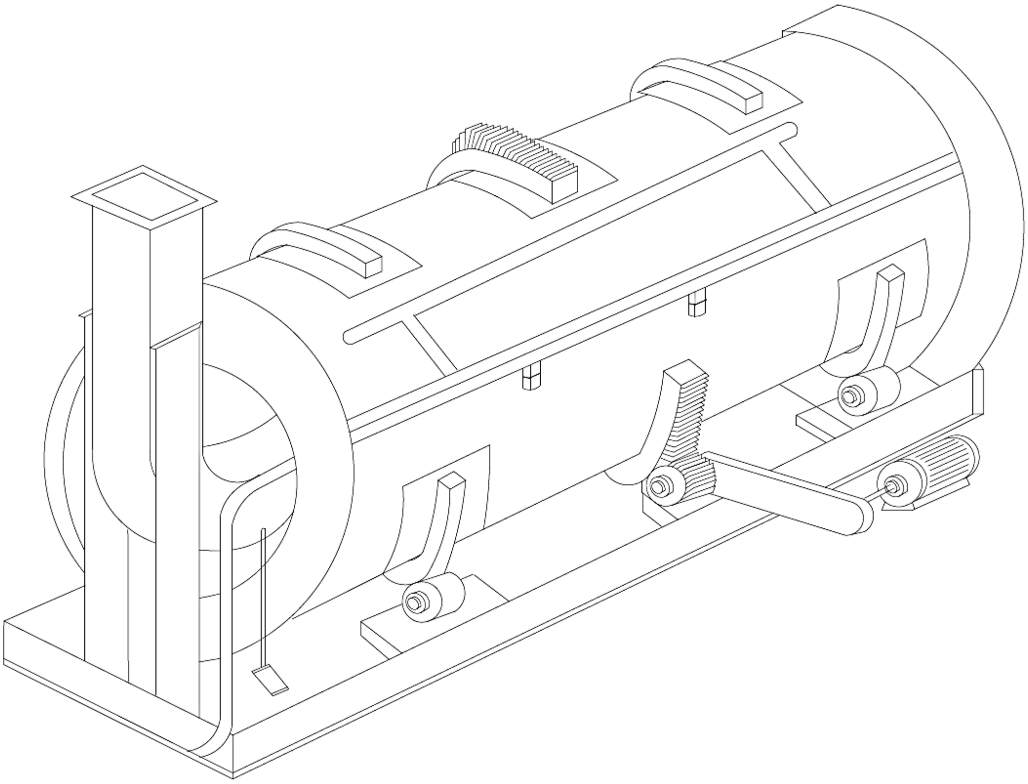
Date: 27/03/17

INDIVIDUAL EQUIPMENT DESIGN



POST ADDITION ROTARY DRUM MIXER AND SPRAY NOZZLES

GROUP NUMBER CE1

NAME Joseph Smith

SUPERVISOR Dr. S. J. Anthony

Date Submitted 27/03/17

EXECUTIVE SUMMARY

This report provides the technical design of a rotary drum mixer and auxiliary equipment for the addition of liquid and powder additives to a base detergent powder. The mass flow inlet of material into the mixer is 8746.24kghr^{-1} with a bulk density of 850kgm^{-3}. The mixer provides a volume of 2.41m^3 based on a design loading 0.15 and residence time of 120s. The drum tilt angle required for this residence time is 1.5^o. The drum has a diameter of 0.8m and a length 4.8m based on a scaling factor of 6:1. The design rotational speed of the mixer is 2.7 rads^{-1} (26rpm) and requires an applied torque of 623Nm. The motor power sized to deliver this is 2.5kW.

The liquid additive consists of perfume oil delivered at a volumetric flow rate of 1.27x10^{-5}m^{3}s^{-1}. This additive is sprayed into the drum with 2 even flat spray nozzles (Model No: QVVA – SS650067) with an individual nozzle flow rate of 3.8x10^{-4}m^{3}min^{-1} delivered at a pressure of 594kPa and a spray angle of 75^o. The nozzles are attached to a sprayer bar with a working distance of 0.4m and a spray width of 0.615m. The nozzles are sited at positions 1.31m and 3.21m along the drum length (4.8m). The corresponding powder velocity required for drop controlled nucleation is 0.74ms^{-1} based on a dimensionless spray flux of 0.07 and a liquid droplet size of 300μm. The liquid is delivered from a #304 stainless steel pipeline 13.2m in length and 9.53mm in diameter by a centrifugal pump with a total power of 120W.

The detergent base powder is stored in a conical mass flow hopper with a volume of 17m^3. This unit has a diameter of 1.94m and a total height of 5.6m. In order to ensure mass flow and prevent arching, the hopper half angle should be designed at a 15.5^o to the vertical with a minimum outlet diameter of 0.088m. The powder is transported to the drum by means of a belt conveyor with a cross sectional area of 9.13x10^{-3}m^2 with a belt width of 0.457m (18in) and at a speed of 0.25ms^{-1}. The total cost for the mixer, auxiliary items and installation (including instrumentation etc.) is €146,000.00.

ACKNOWLEDGEMENTS

I would like to express my gratitude to the project supervisor, Dr S J. Anthony, for his patience and support throughout this project. Despite his own work schedule, he was very gracious with his time for the group and myself during the individual design. I would also like to thank Professors York and Bayly for their deep insight and experience that I could draw from throughout this project. They were also very accommodating in their time and provided the starting point for many learning opportunities throughout this project.

Table of Contents

PAGE COUNT: 44 Pages. Pages 11-60, excluding Pages 43-47 (Engineering Drawings).

NOMENCLATURE

Symbol	Description	Units
$\dot{A}$	Powder Flux through Spray Zone	m^2s^{-1}
$\dot{a}$	rate of production of covered area in spray zone	m^2s^{-1}
μ	Viscosity	Pas
μ_w	Coefficient of Friction	Dimensionless
A	Coefficient A	Dimensionless
A_b	Powder Area formed from Idlers	m^2
A_c	Cross Sectional Area	m^2
A_s	Powder Area formed from Surcharge Angle	m^2
A_t	Total Area formed by Detergent Powder Heap	m^2
b	Belt width	m
B	Coefficient B	Dimensionless
C_A	Capital cost at reference capacity	$
C_B	Capital cost at design capacity	$
C_v	Factor to account for weight of attachments	Dimensionless
D	Diameter	m
D_c	Critical Outlet Diameter	m
$d_{droplet}$	Mean Droplet Diameter	m
D_i	pipe inner diameter	m
$D_{i,opt}$	Economic Pipe Diameter	m
d_{powder}	Mean Particle Diameter	m
Fr	Froude Number	Dimensionless
g	Acceleration due to gravity	ms^{-2}

$H(\Theta)$	constant value dependent on the hopper geometry	**Dimensionless**
h_{eq}	Head Loss due to process equipment	**m**
h_f	Frictional Losses	**m**
h_m	Minor Losses	**m**
H_{max}	Maximum Bed Height	**m**
h_{pump}	Total Head requirement for Pump	**M**
L_{bd}	Powder Bed Depth	**m**
L_{drum}	Drum Length	**m**
L_e	Equivalent Length	**m**
m	Mass	**Kg**
N_h	Hydraulic Power	**W**
N_m	Motor Power	**W**
N_p	Total Power	**W**
N_s	Rotational Speed	**$revmin^{-1}$**
P	Pressure	**Pa**
R	Radius	**m**
S_A	Reference capacity of Unit	**m^3**
S_B	Design Capacity of Unit	**m^3**
S_g	Specific Gravity	**Dimensionless**
T	Torque	**W**
v	Velocity	**ms^{-1}**
v_{bs}	Belt Velocity	**ms^{-1}**
v_L	Linear Velocity	**ms^{-1}**
v_p	Powder Velocity	**ms^{-1}**
W	Spray Width/Height of Powder Fall	**m**

W_m	Weight of Material inside vessel	**N**
W_v	Weight of Vessel based on cylindrical geometry	**N**
X	Drum Loading/Fill Level	**Dimensionless**
Z	Vertical Position	**M**
α	Kinetic Energy Correction Factor	**Dimensionless**
α_s	Angle of Surcharge	**Degrees**
β	Troughing Angle	**degrees**
η	Efficiency	**Dimensionless**
θ	Spray Angle	**Degrees**
λ	Drum Tilt Angle	**degrees**
ρ	Density	**kgm^{-3}**
ρ_b	Bulk Density	**kgm^{-3}**
σ_{crit}	Critical unconfined yield stress	**Pa**
σ_w	Design Stress of Material	**Nmm^{-2}**
τ	Mean Residence Time	**s**
Φ	Static Angle of Repose	**degrees**
Φ_f	Friction Factor	**Dimensionless**
Φ_w	Wall Friction Angle	**Degrees**
ψ_a	Spray Flux	**Dimensionless**
ω	Angular Velocity	**$rads^{-1}$**

TERMINOLOGY

DEM	Discrete Element Method
Fr	Froude Number
LAS	Linear Alkyl Benzene Sulphonate
MOC	Materials of Construction
OEM	Original Equipment Manufacturer
PLC	Programmable Logic Controller
PSD	Particle Size Distribution
TEFC	Totally Enclosed, Fan-Cooler

1 INTRODUCTION

1.1 Project Brief and Selected Equipment

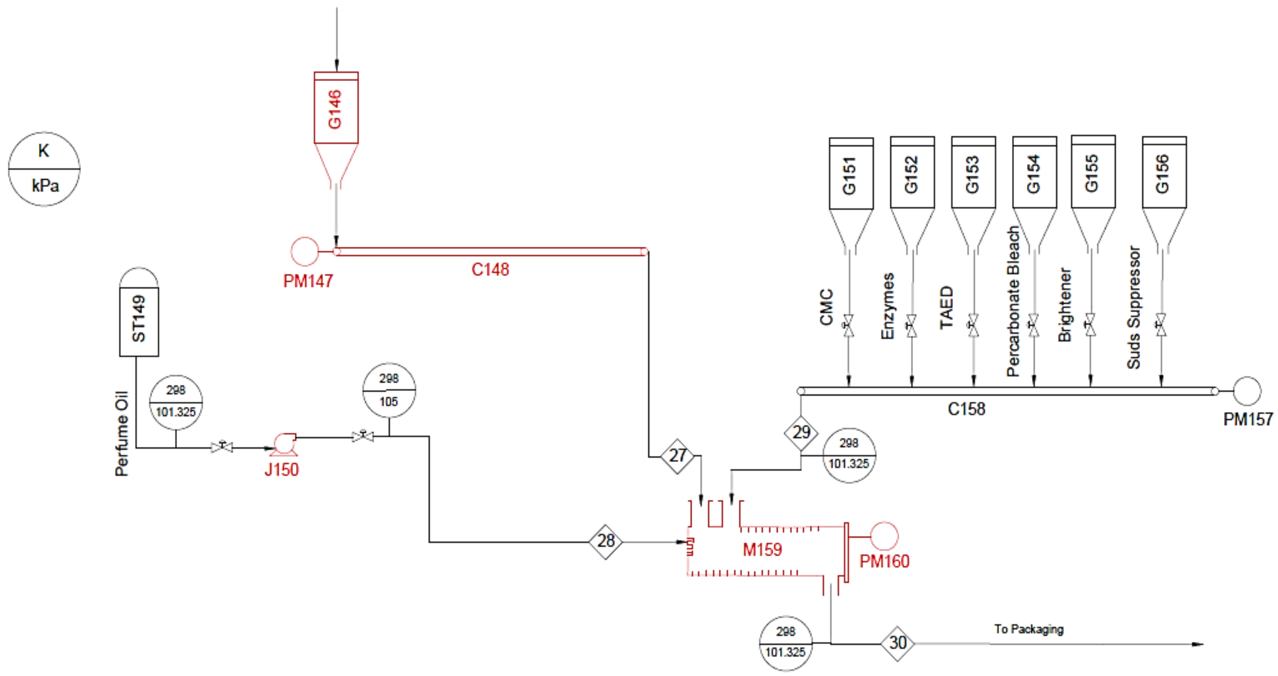
The Thomas Hedley Company have commissioned a technical proposal for the manufacture of a new compact detergent powder in the ‘Dazzle’ brand portfolio, marketed towards Tier 1 consumers in Germany, Austria and Switzerland. The formulation specification for this compact powder is shown in Table 1.1.

Table 1.1. *Formulation Specification for the Dazzle compact detergent powder*

Key Ingredients	**Formulation Level % Active Ingredient**	**Additives and Minor Ingredients***	**Formulation Level % Active Ingredient**
Alkyl Sulphate	5	Brighteners	0.5
Linear Alkyl Benzene Sulphonate	10	CMC	0.5
Non-ionic Surfactant	4	Enzymes	0.5
Zeolite	25	Perfume Encapsulates	1.0
Percarbonate Bleach	12	Perfume Oil	0.5
Sodium Carbonate	18	Polycarboxylate	2.0
Sodium Sulphate	Balance	Sud-suppressor	0.5
Additives/Minors*	8.5	TAED	3.0
TOTAL	100	**TOTAL**	8.5

The proposed plant will produce $6x10^{8}kgyr^{-1}$ of powder with a bulk density of $850kgm^{-3}$ and will be substantially ‘free-flowing, white, dust free and quick to dissolve’. The plant will be sited in Zdemyslice in the Czech Republic and will require flexibility in the equipment design for formulation changes. The processing route selected is a non-tower agglomeration process. The initial stage of the process is dry neutralisation of LAS acid with excess sodium carbonate in a high shear mixer to form a neutralised LAS product. This product is then blended with other major detergent ingredients in a medium shear agglomerator before being dried and cooled in a fluidised bed. This base powder is classified to a design particle size range of 400-600µm in a parallel wire screen with significantly coarse particles reduced in a ball mill. Fines and coarse material outside of the size range are recycled into a storage hopper and fed back into the agglomerator to minimise waste from the plant. The final stage of the process is blending of the powder additives and spraying of liquid additive onto the base powder in a rotary drum. The complete process route and material flow rates are shown in Figure 1.2 (and attached as A3).

The primary objective of this work is the design of the post addition mixer and spray nozzle attachments for the liquid additive. The design work will also include the necessary fluid motive device and pipeline design for the liquid additive (perfume oil) in addition to a surge hopper for the base powder and a conveyor used to transport this material to the mixer. The equipment selected is shown in Figure 1.1 and the materials flows for this stage are shown in Table 1.2.



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<tr><td colspan="2" rowspan="2">UNIVERSITY OF LEEDS

PEME3000 DESIGN PROJECT
INDIVIDUAL EQUIPMENT DESIGN</td><td colspan="4">LEGEND</td></tr>
<tr><td>C</td><td>Conveyor</td><td>MI</td><td>Mixer</td></tr>
<tr><td colspan="2"></td><td>G</td><td>Hopper</td><td>PM</td><td>Motor</td></tr>
<tr><td>VERSION 03</td><td>Author:
JOSEPH SMITH</td><td>J</td><td>Pump</td><td>ST</td><td>Storage Tank</td></tr>
</table>

Figure 1.1 *Post Addition stage of the Detergent Process. Items to be designed are highlighted in red.*

1.2 Design Basis and Requirements

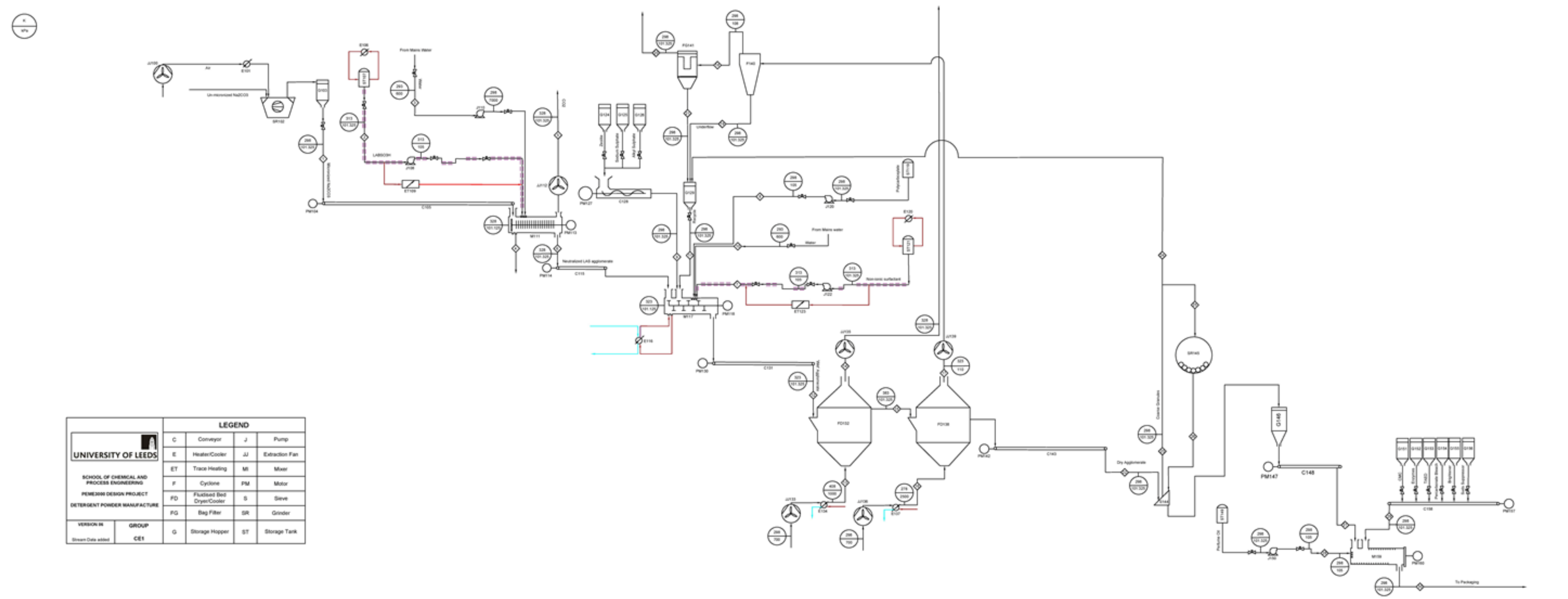
The addition stage blends the base powder received from the fluidised bed cooler with powder additives and the liquid additive in a minimal or low shear mixer. The base powder is dried and cooled to a temperature of 5^{o} prior to blending – this is due to the moisture and temperature sensitivity of the additive components. The mixer design has two primary functions: blending of the powders, and coating the liquid additive onto the powder particles. Poor performance in this secondary aspect will result in a potentially clumping powder with a wide PSD and poor solubility characteristics. The optimal design will be such that the particles are presented to the liquid spray-on from the nozzles over the maximum possible contact area and lightly coat or wet the particle surface before discharging at the outlet of the mixer. An additional concern from the liquids spray on is transient accumulation of powder as the liquid impinges on the mixer wall. The spray angle and position are considered carefully to ensure ideal powder quality and smooth operation (Zoller, 2009). The mixer itself is the last stage in the process prior to packaging of the detergent powder. Potential issues in the design are segregation and size reduction – these can be mitigated through appropriate selection of mixer residence time and design rotational speed. Milling should be avoided at all costs though variance in particle shape, size and characteristics in the mixer cannot be quantified in the scope of this report.

The inlet mass flow rate into the mixer is 8746.24kghr^{-1} into the mixer, which includes the base powder granules, the powder additives and the liquid additive. There are no significant reactions or temperature changes in the system, therefore no auxiliary heating or cooling is required. Prior to the design work, an assessment of the component order of addition was performed as this would significantly influence the spray nozzle selection and drum sizing requirements. It was noted that Polycarboxylate should be moved to the agglomeration stage of the process due to its significant moisture content, and that the suds suppressor should be sourced as a powder rather than a liquid. The mass flows were updated correspondingly to reflect these changes. In terms of equipment design, an additional surge hopper was added for the base powder as this is fundamental in the control strategy for the finishing stage, with a belt conveyor added to transport the material.

Much of the design methodology for the items, in particular the surge hopper and powder bed characteristics, are based on empirical models and powder properties provided in York et al, 2003 and Bayly et al, 2017. Practical design requires knowledge of the powder rheology and characteristics prior to design of the mixer. As a result, assumptions are made and justified to the best available information with magnitude estimations substituting for experimentation (e.g. linear velocity as a model of powder velocity).

***Table 1.2**. Relevant mass flows for the post addition stage*

	STREAM NUMBER (kghr^{-1})							
	27	Ph	28	Ph	29	Ph	30	Ph
NaLAS	856.16	(s)					856.16	(s)
LABSA								
Na_2CO_3	1541.10	(s)					1541.10	(s)
Zeolite	2140.41	(s)					2140.41	(s)
Na_2SO_4	1113.01	(s)					1113.01	(s)
Alkyl Sulphate	428.08	(s)					428.08	(s)
Alkyl Ethoxylate	342.46	(s)					342.47	(s)
Water	370.46	(s/l)	1.31	(l)	13.13	(s)	389.90	(s/l)
Carbon Dioxide								
H_2SO_4								
Misc	36.31				148.66	(s)	184.97	(s)
Air								
Bleach					1027.40	(s)	1027.40	(s)
Enzyme					42.81	(s)	42.81	(s)
Perfume Oil			42.81	(l)			42.81	(s)
Perfume Encapsulates					85.62	(s)	85.62	(s)
Brighteners					42.81	(s)	42.81	(s)
Polycarboxylate	171.23	(l)	171.23	(l)			171.23	(s)
CMC					85.62	(s)	42.81	(s)
TAED					256.85	(s)	256.85	(s)
Suds					42.81	(s)	42.81	(s)
Totals	6999.24		537.54		1659.20		8746.24	



STREAM NUMBER (kghr-1)

	1	Ph	2	Ph	3	Ph	4	Ph	5	Ph	6	Ph	7	Ph	8	Ph	9	Ph	10	Ph
NaLAS											684.93	(s/l)								
LABSA			801.99	(l)			801.99	(l)			160.40	(l)								
Na2CO3	1680.33	(s)									1567.14	(s)								
Zeolite																	2140.41	(s)		
Na2SO4											12.11	(s)					1094.06	(s)		
Alkyl Sulphate																	428.08	(s)		
Alkyl Ethoxylate													342.47	(l)						
Water	25.72	(s)	12.53	(l)	8.02	(l)	20.55	(l)			65.50	(s/l)			279.40	(l)	50.94	(s/l)	500.00	(l)
Carbon Dioxide	8.57	(g)							55.56	(g)										
H2SO4			8.35	(l)			8.35	(l)												
Misc			12.53	(s/l)			12.53	(s/l)			12.53	(s/l)					23.78	(s/l)		
Air																				
Bleach																				
Enzyme																				
Perfume Oil																				
Perfume Encapsulates																				
Brighteners																				
Polycarboxylate															171.23	(l)				
CMC																				
TAED																				
Suds																				
Totals	1714.62		835.41		8.02		843		55.56		2502.60		342.47		450.60		3737.23		500.00	

	11	Ph	12	Ph	13	Ph	14	Ph	15	Ph	16	Ph	17	Ph	18	Ph	19	Ph	20	Ph
NaLAS	201.45	(s)	1057.61	(s)					1057.61				158.64	(s)	150.71	(s)	7.93	(s)		
LABSA																				
Na2CO3	362.61	(s)	1903.71	(s)					1903.71	(s)			285.60	(s)	271.28	(s)	14.28	(s)		
Zeolite	503.61	(s)	2644.04	(s)					2644.04	(s)			396.60	(s)	376.78	(s)	19.83	(s)		
Na2SO4	261.88	(s)	1374.90	(s)					1374.90	(s)			206.20	(s)	195.92	(s)	10.24	(s)		
Alkyl Sulphate	100.73	(s)	528.81	(s)					528.81	(s)			79.32	(s)	75.36	(s)	3.97	(s)		
Alkyl Ethoxylate	80.58	(l)	423.05	(l)					423.50	(l)			63.46	(g)	60.28	(g)	3.17	(g)		
Water	36.83	(s/l)	937.06	(s/l)	1095.07	(g)	1624.83	(g)	407.30	(s/l)	2918.35	(g)	2936.66	(g)	17.39	(g)	2919.27	(g)	2918.35	(g)
Carbon Dioxide			10.81	(g)			10.81	(g)												
H2SO4																				
Misc	8.54	(s/l)	44.86	(s/l)					44.86	(s/l)			6.73	(s/g)	6.39	(s/g)	0.34	(s/g)		
Air					36502.20	(g)	36502.20	(g)			97278.36	(g)	97278.40	(g)			97278.36	(g)	97278.37	(g)
Bleach																				
Enzyme																				
Perfume Oil																				
Perfume Encapsulates																				
Brighteners																				
Polycarboxylate	40.30	(l)	211.52	(l)					211.52	(l)			31.73	(l)	30.14	(l)	1.59	(l)		
CMC																				
TAED																				
Suds																				
Totals	1596.54		9136.36		37597.27		38137.84		8595.80		100196.71		101443.30		1184.25		100259.00		100196.71	

	21	Ph	22	Ph	23	Ph	24	Ph	25	Ph	26	Ph	27	Ph	28	Ph	29	Ph	30	Ph
NaLAS	7.93	(s)	898.97	(s)	214.04	(s)	42.81	(s)	171.23	(s)	171.23	(s)	856.16	(s)					856.16	(s)
LABSA																				
Na2CO3	14.28	(s)	1618.15	(s)	385.27	(s)	77.10	(s)	308.22	(s)	308.22	(s)	1541.10	(s)					1541.10	(s)
Zeolite	19.83	(s)	2247.43	(s)	535.10	(s)	107.20	(s)	428.08	(s)	428.08	(s)	2140.41	(s)					2140.41	(s)
Na2SO4	10.31	(s)	1168.66	(s)	276.54	(s)	55.65	(s)	221.23	(s)	222.60	(s)	1113.01	(s)					1113.01	(s)
Alkyl Sulphate	3.97	(s)	449.50	(s)	107.02	(s)	21.40	(s)	85.62	(s)	85.62	(s)	428.08	(s)					428.08	(s)
Alkyl Ethoxylate	3.17	(g)	359.59	(s)	85.62	(s)	17.12	(s)	68.50	(s)	68.50	(s)	342.46	(s)					342.47	(s)
Water	0.92	(g)	399.99	(s/l)	92.62	(s/l)	18.53	(s/l)	74.09	(s/l)	74.09	(s/l)	370.46	(s/l)	1.31	(l)	13.13	(s)	389.90	(s/l)
Carbon Dioxide																				
H2SO4																				
Misc	0.34	(s/g)	38.13	(s)	9.08	(s)	1.82	(s)	7.26	(s)	7.26	(s)	36.31	(s)			148.66	(s)	184.97	(s)
Air																				
Bleach																	1027.40	(s)	1027.40	(s)
Enzyme																	42.81	(s)	42.81	(s)
Perfume Oil															42.81	(l)			42.81	(l)
Perfume Encapsulates																	85.62	(s)	85.62	(s)
Brighteners																	42.81	(s)	42.81	(s)
Polycarboxylate	1.59	(l)	179.79	(l)	42.81	(l)	8.56	(l)	34.25	(l)	34.25	(l)	171.23	(l)					171.23	(l)
CMC																	42.81	(s)	42.81	(s)
TAED																	256.85	(s)	256.85	(s)
Suds																	42.81	(s)	42.81	(l)
Totals	62.33		7349.21		1749.81		349.96		1399.85		1399.85		6999.24		44.11		1702.89		8746.24	

Figure 1.2**. Process Flow Diagram for Detergent Powder Process.* ***NB: A3 copy enclosed in report.

2 CHEMICAL ENGINEERING DESIGN

2.1 Process Requirements and Selection Methodology

The preliminary consideration for the design is a preferred mixer type. The production rate at the outlet of the mixer will be 8746.23kghr^{-1}, which constrains the selection to mixers that operate on a continuous basis. In terms of product quality, the outlet product should maintain the bulk density (850kgm^{-3}) and the free flowing nature of the powder. In low shear applications, for free flowing powders, the typical mixer types used are conical screw, tumbling and ribbon mixers. Convective mixers such as conical screws and ribbon mixers use impellers to blend the powder feeds, though this can lead to particle attrition and breakage of agglomerates (Harnby et al. 1992). Additionally, the increased number of mechanical components in these mixers can result in more frequent cleaning and higher maintenance cost (Paul et al, 2004).

Tumble mixers are simple designs that rotate on a central axis, mixing the feed powders via diffusion. The most common continuous tumble mixer design is a horizontal cylinder or drum, which is set on rollers to provide the axial movement and trunnions to allow the drum angle of inclination to be changed. The advantages of tumble mixers are their low cost, short cleaning times and long operating life due to lack of moving parts (Paulsworth, 2017). In the current application, a rotating cylindrical drum provides the low shear environment necessary for the additives and continuous operation at low cost. The downsides to the design is the tendency of powders to demix (or segregate) after a given residence time, and is therefore not suitable for blending powders with differences in density (Weinkotter et al, 2000). Figures 2.1 and 2.2 provide a summary of mixer designs and performance characteristics under different conditions.

	Size reduction	Sensitive to differences in density	Sensitive to differences in particle size	Dust formation	Moisture can be added	Possibility of pressure or vacuum	Possibility of heating or cooling	Flexible	Easy to empty	Easy to clean	Rapid wear	Large space required	High power consumption	Rapid mixing	Possibility of continuous mixing
Rotating drum		+	±		+	+	±	±	+	+	-	+	-	-	+
Double cone		+	±		+	+	±	+	+	+	-	+	-	-	-
Twin shell mixer parallel to axis		+	±		+	+	+	+	+	+	-	+	-	±	-
Twin shell mixer perpendicular to axis		+	±		+	+	+	+	+	+	-	+	-	-	-
Ribbon mixer	±	-	-	+	+	+	+	±	±	-	±	-	-	±	+
Orbiting screw	±	±	±	+	±	±	±	±	+	±	±	-	-	±	-
High speed impeller cylindrical vessel	+	-	-	+	+	+	+	+	±	±	+	-	+	+	-
High speed impeller biconical vessel	+	-	-	+	+	+	+	+	±	±	+	-	+	+	-
High speed impeller spherical vessel	±	-	-	+	+	+	+	+	±	±	+	-	±	+	±
Pan mixer	+	±	-	+	+	±	+	±	+	±	±	-	±	±	±
End runner mill	+	±	±	±	±			±	-	-	+	±	±	-	±
Edge runner mill	+	-	±	+	±			±	-	-	+	±	+	-	±
Muller	+	-	±	+	±			±	-	-	+	±	+	+	±
Air mixer	+	+	+	+		±	+	-	±	±	+	±	±	+	+

Figure 2.1. *Characteristics of different mixer designs; + = yes, - = no, ± = satisfactory (Betrand, 1991)*

Type of mixer	*Batch or continuous*	*Main mixing mechanism*	*Segregation (suitability for ingredients of different properties)*	*Axial mixing*	*Ease of emptying*	*Tendency to segregate on emptying*	*Ease of cleaning*
Horizontal drum	B	Diffusive	Bad	Bad	Bad	Bad	Good
Lödige mixer	B	Convective	Good	Good	Good	Good	Bad
Slightly inclined drum	C	Diffusive	Fair	Bad	Good	Good	Good
Steeply inclined drum	B	Diffusive	Bad	Good	Bad	Bad	Good
Stirred vertical cylinder	B	Shear	Bad	Good	Good	Bad	Good
V mixer	B	Diffusive	Bad	Bad	Good	Bad	Good
Y mixer	B	Diffusive	Bad	Bad	Good	Bad	Good
Double cone	B	Diffusive	Bad	Bad	Good	Bad	Good
Cube	B	Diffusive	Poor	Good	Good	Bad	Good
Ribbon blender	B	Convective	Good	Slow	Good	Fair	Fair
Ribbon blender	C	Convective	Good	Fair	Good	Good	Fair
Air jet mixer	B	Convective	Fair	Good	Good	Good	Fair
Nauta mixer	B	Convective	Good	Good	Good	Good	Bad

Figure 2.2. *Mixer Characteristics for free-flowing powders (Harnby, 1992)*

In Figure 2.1, the options for a continuous operation are ribbon blender mixers and slightly inclined drums. The preferred option is the slightly inclined drum, due to their popularity in industrial manufacture of detergents, lower space requirements for installation and the fact that rapid mixing is not the primary function of this stage (Paul et al, 2004).

2.2 Drum Mixer Design Method

2.2.1 Design Guidelines and Assumptions

The initial stage of the design will determine the volume and corresponding dimensions of the drum. An accurate design would require particle modelling through DEM or Population Balance models to determine the mean particle residence time. In industry, scale up from a manufacturer would be provided after pilot scale studies with the coating agents and base powder had been performed (Salman et al, 2007). In lieu of these, a selection of assumptions about the system are made with appropriate scaling factors.

2.2.2 Drum Volume

The volume of the drum can be found as a function of the mass flow rate, residence time and volumetric fill level. In order to convert the mass flow into a volumetric flow rate, a constant density is assumed of all material entering and leaving the mixer for both liquid and solid flows. As part of this assumption, the liquid additive should not significantly impact the powder properties of the detergent. Other factors that could affect the bulk density and particle properties, such as particle attrition and breakage, are assumed negligible within the desired flow regime. The volumetric flow rate is given by (Salman et al, 2007):

$$\dot{V} = \frac{\dot{m}}{\rho_b} \tag{2.1}$$

Where: $\dot{V}$= volumetric flow rate of drum (m^3s^{-1}), ρ_b = bulk density of powder (kgm^{-3}),

The mean residence time of a particle within the drum is governed by the drum length, tilt angle and powder properties. An appropriate residence time must be selected to ensure that the particles are sufficiently coated by the liquid additive without leading to overmixing and particle segregation. An appropriate value specified from industrial practice of coating detergent powder is 120 seconds (Davies et al, 1985), confirmed by a general guideline range of 120 – 300s (Litster et al, 2004). Given the low level of liquid addition, it is preferred to use a residence time at the lower bound of this range. The final consideration, the fill volume or drum loading, is selected based on the desired flow regime for the drum. The typical loading of the drum is in the range of 10%-30%, with 15% selected in order to satisfy the theoretical requirement for a cataracting regime (Litster et al, 2004). This generates the following equation (Salman et al, 2007):

$$\frac{\dot{V} \cdot \tau}{X} = V \tag{2.2}$$

Where: X = volumetric fill level, τ = mean residence time (s), V = volume of drum under design loading (m^3)

Based on these assumptions, the volume of the drum under design loading may then be found:

$$\dot{V} = \frac{\dot{m}}{\rho_b} = \frac{8746.23kghr^{-1}}{850kgm^{-3}} \cdot \frac{1hr}{3600s} = 2.86x10^{-3}m^3s^{-1}$$

$$V_A = \frac{\dot{V} \cdot \tau}{0.15} = \frac{\left(2.86x10^{-3} m^3 s^{-1}\right)\left(120s\right)}{0.15} = 2.29m^3$$

2.2.3 Drum Diameter and Length

The diameter and length of the drum can be related to the volume by assuming that the drum is a perfect cylinder. In practice, the inclusion of sprayer bars, weirs and other mechanical components will impact the actual available volume of the drum. However, in this design the drum fill level is sufficiently low that these components should have negligible effect on the flow regime of the powder bed. This results in equation **(2.3)**.

$$V = \pi R^2 \cdot L_{drum} \quad \textbf{(2.3)}$$

Where: V = drum volume (m^3), R = radius of the drum (m), L_{drum} = length of the drum (m)

The length of the drum should be sufficiently long for installation of multiple nozzles with non-overlapping spray coverages (Salman et al, 2007). Additionally, the drum should accommodate a zone at the inlet and outlet of the mixer where no spraying occurs to prevent product accumulation and to allow for product 'aging'. This is typically given as 15% of the drum length on either end of the drum (Zoller, 2009). Guidelines for drum mixers suggest an aspect ratio in the range 2 – 10 times the diameter (Litster et al, 2004) depending on the mass of liquid additive. In this design, an aspect ratio of 6:1 (Length to Diameter) is selected to ensure the drum has the flexibility to accommodate the possibility of additional spray nozzles. This aspect ratio is also readily available from suppliers (Munson Machinery Company Inc, 2016). An appropriate value for the radius and hence the length of the drum is based on the desired flow regime of the powder within the drum.

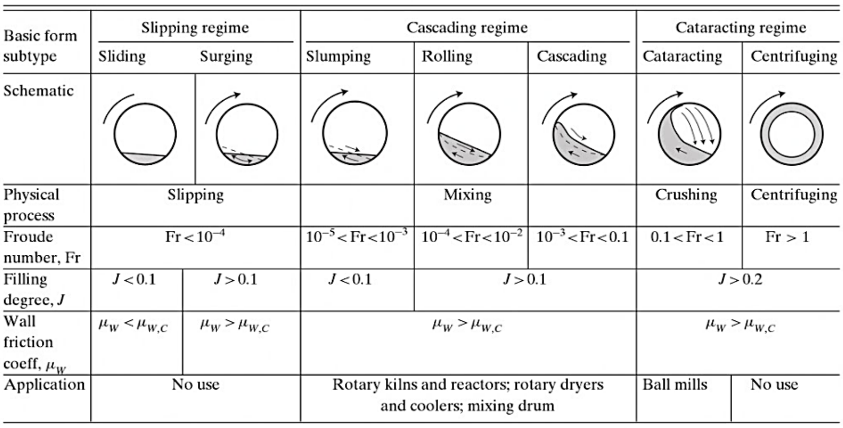
2.2.4 Flow Regimes and Rotational Speed

In a rotary drum, the Froude Number is a ratio of the centrifugal force of the drum (that pushes the powder to the drum wall sides) to the gravitational force which causes tumbling of the powder back to the centre base of the drum (Litster, 2016). The behaviour of the granules in a rotating drum can be broadly classified into three main flow regimes that are a function of the Froude Number **(2.4)**, the internal wall friction and filling degree (Mellmann, 2001). These are the *slipping*, *cascading* and *cataracting* regimes, with differentiation into sub categories of these main regimes (Figure 2.3).

$$Fr = \frac{\omega^2 R}{g} \quad \textbf{(2.4)}$$

Where: Ns = angular velocity of the drum ($rads^{-1}$), R = radius of the drum (m), g = acceleration due to gravity ($9.81ms^{-2}$)

In the *slipping* regime, the powder bed moves uniformly with little mixing of particles and a relatively static surface. This regime occurs at Froude Numbers ($Fr < 10^{-4}$) with sub types of sliding and surging regimes, which are different by the volumetric fill level of the drum (either greater or less than 0.1) and the dynamic friction coefficient between the wall and the powder ($\mu_W < \mu_{W,C}$). In the rotary drum, the optimal design will result from the spray of individual droplets onto single particles, ensuring an even coverage of powder without overwetting. The slipping regime presents only the surface layer of particles which would likely result in only a superficial coating of the powder. The *cascading* regime occurs over a range of $10^{-5} < Fr < 0.1$ where the particles tumble from the apex to the low point of the powder bed. The sub types within this category are *slumping*, *rolling* and *cascading*. In the slumping regime, the particles avalanche as they build up at the apex of the powder bed before falling to the bottom. In the rolling regime, the particles continuously move down the surface of the bed in two primary modes: a well-mixed cascade of particles and a slow uniform movement of particles. For large Froude numbers, a cascading regime is observed where the powder bed arches at the apex of the drum before falling to the bottom. This regime provides excellent mixing of the particles in a dynamic moving powder bed, so is commonly used for drum granulators (Marshall et al, 2014).



Basic form	Slipping regime		Cascading regime			Cataracting regime	
subtype	Sliding	Surging	Slumping	Rolling	Cascading	Cataracting	Centrifuging
Schematic							
Physical process	Slipping			Mixing		Crushing	Centrifuging
Froude number, Fr	$Fr < 10^{-4}$		$10^{-5} < Fr < 10^{-3}$	$10^{-4} < Fr < 10^{-2}$	$10^{-3} < Fr < 0.1$	$0.1 < Fr < 1$	$Fr > 1$
Filling degree, J	$J < 0.1$	$J > 0.1$	$J < 0.1$	$J > 0.1$		$J > 0.2$	
Wall friction coeff, μ_W	$\mu_W < \mu_{W,C}$	$\mu_W > \mu_{W,C}$	$\mu_W > \mu_{W,C}$			$\mu_W > \mu_{W,C}$	
Application	No use		Rotary kilns and reactors; rotary dryers and coolers; mixing drum			Ball mills	No use

***Figure 2.3**. Different flow regimes for a rotary drum, with controlling factors (Marshall et al, 2014)*

At higher rotational speeds, and corresponding Froude Numbers > 0.1, *cataracting* and *centrifuging* regimes are observed. In the cataracting regime, the centrifugal force of the drum is sufficiently high to carry the particles to the top of the drum, at which point gravitational forces dominate and the powder falls to the base of the drum. This generates a falling curtain of particles where surface area is maximised. If the frictional forces are sufficiently high (or $Fr \geq 1$), the particles will remain attached to the drum wall and centrifuge where little mixing occurs as the particles act as a solid body (Marshall et al, 2014). Of the flow regimes shown in Figure 2.3, the cataracting regime offers the maximum particle contact surface area as they are flung into the free space of the drum. The liquid additive can then be sprayed in this falling powder region where the powder flux is high, mitigating the risk of granule agglomeration and ensuring even coverage (Salman et al, 2007).

The Froude Number may be used as a convenient approximation of the flow regime and predictor of particle behaviour for different radii and rotational speeds. The limitation of scaling up by this method is that the powder properties are largely not considered. Typical Froude Numbers for a cataracting regime occur in the range of 0.3 – 0.5 for detergent powders, with 0.3 selected as the optimal value (Litster, 2016). A general scale up relationship can then be established on rearrangement of equation **(2.4)**:

$$N_s = \frac{\sqrt{0.3 \cdot 9.81 ms^{-2}}}{\sqrt{R}} \quad \textbf{(2.5)}$$

NB: Rotational speed is determined in rads^{-1}, and converted to rpm to compare against the design guidelines

The final consideration before calculation of the drum dimensions is the anticipated rotational speed. Using general design guidelines and patent literature, it is suggested that the rotational speed be in the range of 2 – 30rpm (Callaghan Jr, 1996). An initial value estimate for the drum radius is assumed and iterated to satisfy the volume requirements determined from equation **(2.2)**. This radius is given as 0.39m, with a corresponding length of 4.68m based on the aspect ratio. In a practical design and purchasing of a drum, such specific values are rarely required. The radius is therefore rounded to 0.4m with a corresponding length of 4.8m. The added benefit of these dimensions is that the volume of the drum is slightly oversized, which builds flexibility into the design. The rotational speed and new drum volume are then determined:

$$. \ N_s = \frac{\sqrt{0.3 \cdot 9.81 ms^{-2}}}{\sqrt{R}} = \frac{\sqrt{0.3 \cdot 9.81 ms^{-2}}}{\sqrt{0.4}} = 2.71 rads^{-1}$$

$$RPM = \omega \cdot \frac{60}{2\pi} = 2.74 rads^{-1} \cdot \frac{60}{2\pi} = 25.9 rpm$$

$$V = \pi(0.4m)^2 \cdot \frac{(0.4m \cdot 6)}{0.15} = 2.41 m^3$$

$$\therefore D = 0.8m \therefore L_{drum} = 4.8m$$

2.2.5 Characterization of the Powder Bed

The spraying zone within the cataracting regime covers the region of powder fall from the apex of the powder bed to a theoretical point of powder settling at the mid chord position. In order to determine the width of this spray zone, a selection of empirical models are used from York et al, 2003 for this theoretical design. The maximum height of the powder bed H_{max} is defined as the vertical distance between the lowest point of the powder bed and the point at which the powder detaches from the drum wall (Figure 2.4). Similarly, the terminus of the powder fall is taken as the Bed Depth at the mid-chord position (L_{bd}). These two values may then be used to determine the contacting spray width of the nozzles, and hence the spray angle.

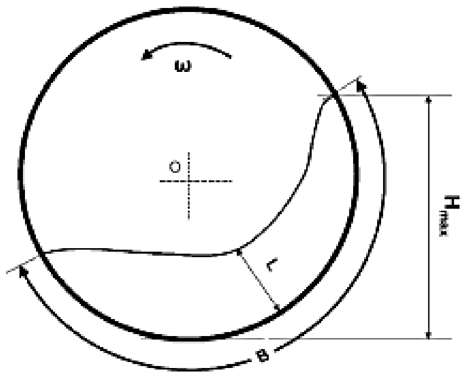


***Figure 2.4**. Geometric parameters of the drum, where: ω = rotational speed, Hmax = apex of powder bed to base of powder bed, L = bed depth, B = tangential contact length between bed and drum wall (York et al, 2003)*

In York et al, 2003, the values of H_{max} and L_{bd} can be approximated from a best fit relationship applied to glass ballotini at different particle diameters and rotational speeds. These are given in equations **(2.6)** and **(2.7)** respectively (York et al, 2003):

$$\left(\frac{H_{\max}}{D}\right)\left(\frac{D}{d_{powder}}\right)^{-0.12} Fr^{-0.06} X^{-0.25}\left(\frac{\tan\phi_w}{\tan\phi}\right)^{0.5} = 0.55 \pm 0.04 \qquad \textbf{(2.6)}$$

$$\left(\frac{L_{bd}}{D}\right)\left(\frac{D}{d_{powder}}\right)^{-0.04} Fr^{-0.1} X^{-0.6}\left(\frac{\tan\phi_w}{\tan\phi}\right)^{0.5} = 0.42 \pm 0.04 \qquad \textbf{(2.7)}$$

The relationships are valid over a Froude Number range of 0.015 - 0.56, which makes them applicable in this design. However, there are notable limitations within the model due to their empirical derivation. In particular, these models were derived from experiments using glass ballotini of roughly uniform particle shape and narrow size distribution. However the base detergent powder and additives are in practicality of non-uniform shape, and have different rheological properties. Similarly, the material properties of the particles are liable to change in the cataracting regime rather than remain uniform throughout their movement in the drum. The effect of aerodynamic forces such as drag on the particles are also not considered. The experimental work was also carried out in a batch scale drum with no liquid addition, which deviates from the current system which operates on a continuous basis with the addition of liquid onto the powder (York et al, 2003). In spite of these limitations, the model provides the best model of the powder bed without undertaking detailed pilot scale work or simulations.

The angle of repose of the detergent powder and the coefficient of friction of the powder against the drum wall are taken from Bayly et al, 2017. These values are 33° and 0.42, respectively. The wall friction angle can then be determined from the coefficient of friction by equation **(2.8)**. The average particle diameter selected is 500µm, which is based on a typical median particle size (Zoller, 2009) (Mort III et al, 2004). This data is summarised in Table 2.1.

$$\phi_w = \tan^{-1}\left(\mu_w\right) \qquad \textbf{(2.8)}$$

Where: μ_w = coefficient of friction between wall and powder

$$\phi_w = \tan^{-1}\left(0.42\right) = 22.8^o$$

__Table 2.1__. Parameters used for estimation of Bed Height (York et al, 2003) (Bayly et al, 2017)

Property of Glass Beads	Symbol	Value	Reference
Particle Diameter	d_p	500µm	Mort III et al, 2004
Fill Level	X	0.15	Litster et al, 2004
Froude Number	Fr	0.3	Litster, 2016
Static Angle of Repose	Φ	33^o	Bayly et al, 2017
Wall Friction Angle	$Φ_w$	22.8^o	Bayly et al, 2017

The values of H_{max} and L_{bd} can then be found:

$$\left(\frac{H_{\max}}{0.8}\right)\left(\frac{0.8}{500x10^{-6}m}\right)^{-0.12}0.3^{-0.06}0.15^{-0.25}\left(\frac{\tan 22.8^o}{\tan 33^o}\right)^{0.5}=0.55\pm0.04$$

$$\therefore H_{\max}=0.77m$$

$$\left(\frac{L_{bd}}{0.8m}\right)\left(\frac{0.8m}{500x10^{-6}m}\right)^{-0.04}0.3^{-0.1}0.15^{-0.6}\left(\frac{\tan(22.8^o)}{\tan(33^o)}\right)^{0.5}=0.42\pm0.04$$

$$\therefore L_{bd}=0.153m$$

2.2.6 Drum Angle

The mean residence time of a particle within the drum can be varied as a function of the drum angle of inclination, which is measured to the horizontal. In industrial practice, this tilt angle typically ranges from 0 – 10^o depending on the residence time requirement for the drum. In this design, the residence time of the particle is dictated by the length of the cascade path (Figure 2.5). There are a selection of empirical, mechanistic and compartment models that may be used to estimate the residence time of a particle within the drum. Empirical models such as Sullivan, Perry and Chilton and Friedman and Marshall are some of the simplest models available which are derived from rotary kilns. These models are exceptionally limited and tend to give the widest variation in residence time predictions, as they only consider drum properties and not the properties of the powder material (Stephane, 2015). Since these models are poor predictors of residence time, they are not included in the main design though a magnitude assessment was performed and is included in Appendix A. Mechanistic and compartment models provide better predicative results but are typically system-specific and require considerably greater knowledge about the operating parameters than is available in this design (Stephane, 2015).

The theoretical model used to estimate the drum angle is provided by Saeman. In this model, a single granule rotates at the same rotational speed as the drum wall before cascading down the free surface and repeating the movement towards the exit of the drum (Anthony, 2004). The direction of this cascade is inclined to the horizontal of the angle of repose (Figure 2.5). The model is based on a series of assumptions: firstly, that the time taken to cascade down the free surface is negligible compared to the drum rotation, that the bed surface is locally flat and that angle of inclination of the drum is relatively small (Anthony, 2004). In order to derive the residence time equation given by **(2.9)**, the bed surface is assumed as locally flat (dh/dx = 0). In a cataracting regime, this assumption is not valid though the model provides the most accurate estimate of the residence time for this design as it includes powder properties and does not require detailed knowledge of the system.

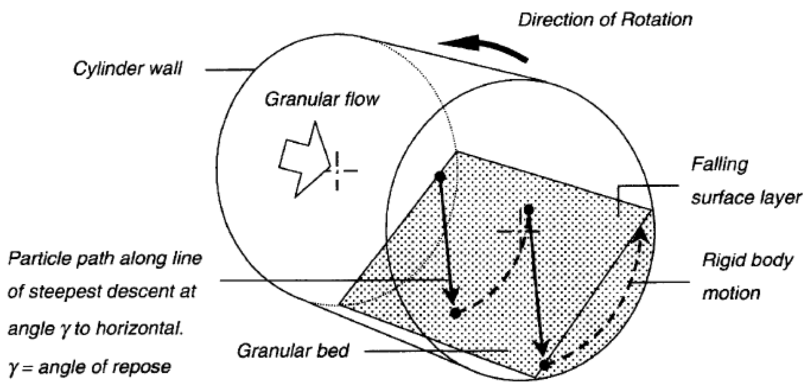


Figure 2.5. *Diagram of granule movement through kiln (Anthony, 2004)*

The mean residence time can be predicted by the following (Anthony, 2004):

$$\tau = \frac{3L_{drum}\sin(\phi)}{4\pi n R \tan\lambda}\left[\frac{\cos^{-1}(1-y)-\left(1-y(y(2-y))^{0.5}\right)}{[y(2-y)]^{1.5}}\right] \quad \textbf{(2.9)}$$

Where: τ = mean residence time of granule, L = length of drum (m), Φ = angle of repose (deg), N_s = rotational speed (revs^{-1}), R = drum radius (m), λ = angle of inclination (deg)

And: $$y = \frac{L_{bd}}{R} \quad \textbf{(2.10)}$$

The drum tilt angle can then be calculated as 1.5^o for a residence time of 120 seconds given from equation **(2.10)**. In practical designs, the drum angle of inclination would be varied manually before the plant begins full operations in order to achieve the desired residence time.

$$y = \frac{L_{bd}}{R} = \frac{0.153m}{0.4m} = 0.382$$

$$120s = \frac{3(4.7m)\sin(33^o)}{4\pi(0.44revs^{-1})(0.39m)\tan\lambda}\left[\frac{\cos^{-1}(1-0.382)-\left(1-0.382(0.382(2-0.382))^{0.5}\right)}{[0.382(2-0.382)]^{1.5}}\right]$$

$$\therefore \lambda = 1.5^o$$

The generic design guidelines have been collated and presented in Appendix C.

2.3 Spray Nozzle Design

2.3.1 Spray Nozzle Selection

The selection of an appropriate spray nozzle is dictated by the fluid properties and the desired spraying performance. Broadly, there are two common methods of atomization for spray nozzles: hydraulic nozzles which use the kinetic energy of a single fluid to break the fluid into droplets and two fluid nozzles which use a separate gas stream to atomize the liquid. In the former case, single fluid nozzles are used for low pressure applications whereas two fluid nozzles are applied for viscous fluid applications. Other methods, such as ultrasonic, are not considered in this design. The liquid additive, perfume oil, has a viscosity of 0.015 Pas and a density of 940 kgm^{-3} at 20^{o}C. Due to the relatively low viscosity of the fluid, it is not necessary to select an air atomizing or two fluid nozzle to atomize the liquid and single fluid nozzles are sufficient (Mayam, 2006) (PNR, 2014).

The secondary consideration for spray nozzles include the spray pattern, spray capacity, the angle of the spray and the desired droplet size. Nozzle patterns can be classified into hollow cone, full cone, flat sprays and solid stream. Of those listed, flat fan nozzles yield a smaller droplet size than full cone nozzles for a uniform spray pattern. Additionally, a non-tapered (even) flat fan nozzle provides minimal overlap and has the advantage of a high aspect ratio which means these nozzles can also be sited in parallel, if the liquid flux from a single nozzle is too high (Spraying Systems Co, 2016). The flat spray provides a simpler model to characterise the liquid spray flux and the degree of particle wetting.

2.3.2 Theoretical Spray Angle

The spray zone of the liquid additive should cover the region where the powder is cascading in the free space of the drum, as it is highly mobile and ensures good distribution of liquid. The distance of this powder cascade may be determined from the maximum bed height (H_{max}) and the bed depth at the mid-chord position (L_{bd}), given by equation **(2.11)**:

$$W = H_{\max} - L_{bd} \tag{2.11}$$

Where: W = distance of powder fall in cataracting flow

If the spray nozzle is assumed to be situated at the axis of rotation of the drum (at the centre), the working distance of the spray is given by the radius R. Taking the powder bed as flat chord across the drum, equation **(2.12)** may be derived to determine the spray angle based on simple trigonometry.

$$\tan\left(\frac{\theta}{2}\right) = \left(\frac{W}{2}\right) \cdot R = \left(\frac{H_{\max} - L_{bd}}{2}\right) \cdot R \tag{2.12}$$

Where: Θ = spray angle

An illustration of this derivation is provided in Figure 2.6.

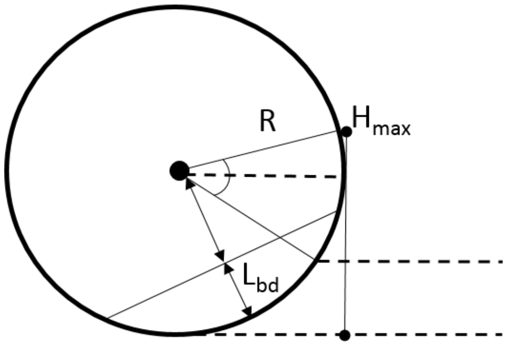


Figure 2.6. *Schematic diagram of parameters used to estimate spray angle for coverage of falling powder.*

The theoretical spray angle may then be determined from equation **(2.12)**.

$$\tan\left(\frac{\theta}{2}\right) = \left(\frac{0.77 - 0.153}{2}\right)(0.4m)$$

$$\tan\left(\frac{\theta}{2}\right) = 0.1234$$

$$\theta = 75.08^{o} \approx 75^{o}$$

This approximation is the maximum spray angle required to cover the powder fall. Common industrial practice typically does not exceed a spray angle of 110^{o} to avoid overlapping spray zones and impinging spray on the drum wall (Zoller, 2009). Similarly, this is only a theoretical spray angle as the actual spray angle will be impacted by the surface tension of the liquid, the working distance from the powder surface and aerodynamic effects. Models to compensate for this real world deviation are beyond the scope of this work, so it is assumed that the theoretical angle is representative of the actual spray angle given the relatively short working distance (0.4m). It is possible to site the sprayer bar and nozzle attachments at a distance further than the radius, which would mitigate potential blockages from particles liberated during the cataracting motion of the drum. However, this may compound the aerodynamic effects acting on the droplets and potentially lead to drifting of the droplets and impinging on the drum wall. This transient accumulation of material on the drum wall should be avoided as it can adversely affect the production and quality of the final stage powder (Zoller, 2009).

2.3.3 Spray Nozzle Pressure and Flow Rate

The selection of an appropriate spray nozzle can now be made from manufacturer catalogues, given the desired spray angle and nozzle type. The volumetric flow rate of perfume oil into the drum is given by equation **(2.1)**, for a mass flow rate of 42.81kghr^{-1} and a density of 940kgm^{-3}.

$$\dot{V} = \frac{\dot{m}}{\rho} = \frac{42.81kghr^{-1}}{940kgm^{-3}} \cdot \frac{1hr}{60\min} = 7.6x10^{-4} m^{3} \min^{-1}$$

In order to achieve this flow rate, two QVVA #303 stainless steel nozzle are selected from Spraying Systems Co. that provide a spray angle of 75^{o} and a water flow rate of 0.37Lmin^{-1} at 6 bar pressure (600kPa). The supplier catalogue is shown in Figure 2.7.

Spray Angle at 3 bar	Quick VeeJet Tip Type						Capacity Size	Equiv. Orifice Dia. (mm)	Flow Rate Capacity (liters per minute)									Spray Angle (°)			
	QSVV	QVVA	QUA	QLUA	QMVV	QPTA			0.4 bar	0.7 bar	1.5 bar	3 bar	6 bar	7 bar	12* bar	15** bar	20 bar	1.5 bar	3 bar	6 bar	15 bar
65°		●					0017	.28	–	–	.047	.067	.095	.10	.13	.15	.17	44	65	77	86
		●					0025	.33	–	–	.070	.099	.14	.15	.20	.22	.25	45	65	77	84
		●					0033	.38	–	–	.092	.13	.18	.20	.26	.29	.34	47	65	76	83
		●					0050	.46	–	–	.14	.20	.28	.30	.39	.44	.51	48	65	75	82
		●					0067	.53	–	.13	.19	.26	.37	.40	.53	.59	.68	50	65	75	81
		●					01	.66	–	.19	.28	.39	.56	.60	.79	.88	1.0	51	65	74	80
		●					015	.81	–	.29	.42	.59	.84	.90	1.2	1.3	1.5	51	65	74	80
	●	●			●		02	.91	.29	.38	.56	.79	1.1	1.2	1.6	1.8	2.0	52	65	73	79

Figure 2.7. *Performance data for flat spray VeeJet Nozzles (Spraying Systems Co. 2016)*

An operating pressure of 600kPa provides the necessary spray angle for powder coverage and it is expected that the mean droplet size will be approximately 300µm for a very fine nozzle head. It is possible to approximate the droplet size of the fluid as a function of its properties such as surface tension and density, though the dependence of droplet size on the fluid is very complex (AspenONE, 2016). As such, a pressure of 600kPa is taken as sufficient for the desired droplet size for perfume oil (Litster et al, 2004). The total capacity or volumetric flow rate of these nozzles is given by:

$$\dot{V} = 2 \cdot \left(0.37L\min^{-1}\right) = 0.74L\min^{-1} = 7.4x10^{-4}m^{3}\min^{-1}$$

However, this flow rate is based on an equivalent flow rate of water with a specific gravity of 1 and must therefore be converted based on the density of perfume oil using equation **(2.13)**(Spraying Systems Co, 2016). Hence, the spray nozzle flow rate for the perfume oil is calculated as 7.63x10^{-4}m^{3}min^{-1}.

$$\dot{V} = \dot{V}_{water} \cdot \frac{1}{S_g^{\ 0.5}} \qquad \textbf{(2.13)}$$

Where: $\dot{V}_f$ = volumetric flow rate of liquid, $\dot{V}_{water}$ = analogous flow rate for water

$$\dot{V} = \dot{V}_{water} \cdot \frac{1}{S_g^{\ 0.5}} = 7.4x10^{-4}m^{3}\min^{-1} \cdot \left(\frac{1}{0.94^{0.5}}\right) = 7.63x10^{-4}m^{3}\min^{-1}$$

As is shown above, the flow rate of the two nozzles at a pressure of 600kPa is slightly higher than the design volumetric flow rate. To satisfy the rate of addition, a relationship between different nozzles capacities as a function of their pressures may be used to determine the actual operating pressure of the nozzles, given by equation **(2.14)** (PNR, 2014).

$$\frac{P_2}{P_1} = \left(\frac{\dot{V}_2}{\dot{V}_1}\right)^2 \quad \textbf{(2.14)}$$

Where: $\dot{V}_2$ = volumetric flow rate of perfume oil for the formulation requirement (m^3s^{-1}), $\dot{V}_1$ = volumetric flow rate of perfume oil at 600kPa (m^3s^{-1})

The actual pressure of the spray nozzle to achieve this flow rate is therefore:

$$P_2 = \left(\frac{\dot{V}_2}{\dot{V}_1}\right)^2 P_1 = \left(\frac{7.6x10^{-4}m^3 \min^{-1}}{7.63x10^{-4}m^3 \min^{-1}}\right)^2 (6bar) = 5.93bar = 593kPa$$

This is the operating pressure of the spray nozzles. In the design of the pipeline and pumping requirements, the pump should overcome any losses in the pipeline and deliver the perfume oil to the nozzles at the calculated pressure energy. It is assumed throughout this calculation that the viscosity and density of the fluid will remain constant within the pipeline and nozzles, and does not account for seasonal variation in temperatures. Due to the low level of perfume oil addition, it would be possible to site the storage vessel within the main body of the plant itself where temperature is relatively constant due to central heating, rather than in an outside tank farm. It is also assumed that the perfume oil acts as a Newtonian fluid, with a linear response to the shearing effects of the pump and nozzles. A summary of the nozzle design requirements are provided in Table 2.2.

Table 2.2. *Summary of Design Information for Spray Nozzles*

Component	Density (kgm^{-3})	Viscosity (Pa.s) @20^{o}C	Spray Angle (degrees)	Flow Rate (m^3min^{-1})	Number of Nozzles	Flow Rate per nozzle (m^3min^{-1})	Nozzle Operating Pressure (kPa)	Model
Perfume Oil	940	0.015	75	$7.6x10^{-4}$	2	$3.8x10^{-4}$	593	QVVA – SS650067

2.3.4 Particle Wetting and Dimensionless Spray Flux

The liquid additive is distributed from the spray nozzles in the form of droplets onto the moving powder granules. The wetting performance of these granules with the liquid additive can influence the particle properties of the detergent powder, with poor wetting resulting in poorer solubility characteristics in the wash and potentially caking or clumping that is in violation of the design specification (Zoller, 2009). In this design, the droplet size is taken as 300µm which is lower than the particle size range of 400-600µm. This leads to coating of the particles from the collision of the granules and the liquid droplets. Conversely, droplet sizes larger than the particle size may form nuclei as an initial stage in agglomeration. In either case, characterisation of the particle wetting and liquid distribution must be performed in order predict the properties of the end stage powder (Salman et al, 2007).

The wetting of particles is broadly classified into three regimes: droplet controlled nucleation, shear controlled and intermediate. In a droplet controlled regime, each droplet wets an individual granule of the powder bed to form a single nuclei. The penetration time of the droplet onto the granule surface is low and droplets minimally overlap. This regime is preferred for the current design, as particle size remains relatively constant and controllable by the rate of liquid flux. The shear controlled regime results in caking of the powder and poor distribution of the liquid additive, which may only be distributed by shearing of the granules. Clearly, this regime should be avoided in the current design. The intermediate regime occurs as a median position between the other two regimes, with some agglomeration depending on the properties of the powder and liquid additives (Litster et al, 2004). These regimes, as a function of the drop penetration time and dimensionless spray flux, are shown in Figure 2.8.

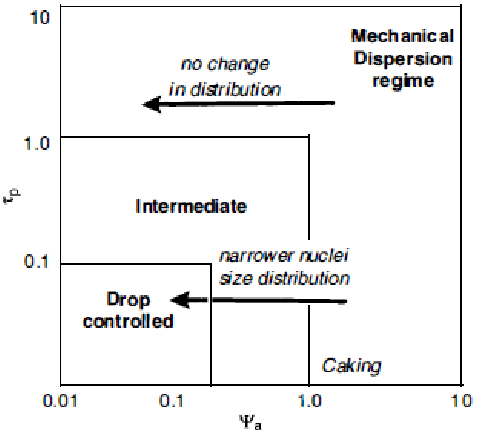


Figure 2.8. *Nucleation regime map. Drop controlled nucleation is achieved with a spray flux and drop penetration time < 0.1 (Salman et al, 2007)*

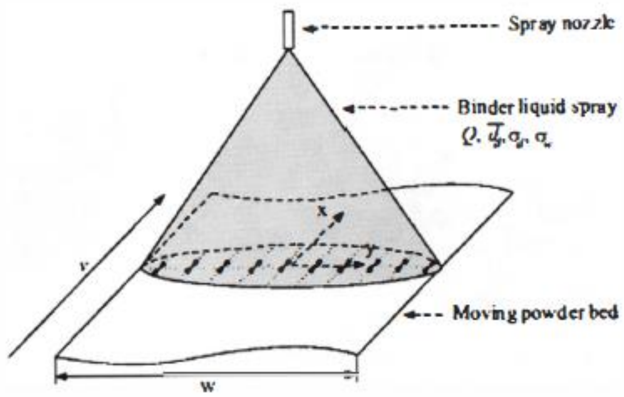


Figure 2.9. *Representation of spray zone in drum mixer for a flat spray (Salman et al, 2007)*

The type of regime and wetting performance is governed by the droplet penetration time and ratio of liquid to solid flux (dimensionless spray flux). The droplet penetration time requires knowledge of the powder surface properties, and is typically determined experimentally through a battery of tests. In this design, methods such as the Washburn test are not available so only the dimensionless spray flux is considered. For a flat spray, the movement of liquid droplets through a powder flux may be approximated by equation **(2.15)** (Salman et al, 2007). This spraying zone is also shown in Figure 2.9.

$$\psi_a = \frac{\dot{a}}{\dot{A}} = \frac{3\dot{V}}{2\dot{A}d_{droplet}} \qquad \textbf{(2.15)}$$

Where: ψ_a = dimensionless spray flux, $\dot{a}$ = rate of production of covered area in spray zone (m^2s^{-1}), $\dot{V}$ = volumetric flow rate of liquid additive (m^3s^{-1}), $d_{droplet}$ = diameter of liquid droplet (m), $\dot{A}$ = powder flux through spray zone (m^2s^{-1})

In practical scenarios, this equation assumes the liquid flux is uniformly distributed over the spray width and does not consider the fact nuclei granules can coalesce even if liquid droplets do not overlap. A low spray flux ($\psi_a << 1$) corresponds to well-dispersed droplets and effective distribution of the liquid additives on the powder bed (Salman et al, 2007). A value close to 1 corresponds to a high rate of flux leading to agglomeration of the granules. In this design, a spray flux of 0.07 is selected based on the regime map provided in Figure 2.8, corresponding to a droplet controlled regime. The droplet size is provided as 300µm from industrial practice and the specifications outlined in the nozzle catalogue (DiSalvo et al, 1971). The powder surface presented to the liquid spray area ($\dot{a}$) must be continually renewed to prevent over wetting of the granules. For a flat spray zone, the powder flux is given by equation **(2.16)** (Salman et al, 2007):

$$\dot{A} = v_p W \tag{2.16}$$

Where: $\dot{A}$ = powder flux through spray zone (m^2s^{-1}), v_p = powder velocity (ms^{-1}), W = width of spray zone (m)

Substitution of equation **(2.16)** into equation **(2.15)** and rearrangement in terms of powder velocity yields the following:

$$v_p = \frac{3\dot{V}}{2\psi_a d_{droplet} W} \tag{2.17}$$

The result of this is to determine whether the desirable conditions within the drum, with the rate of liquid addition, are feasible from the magnitude of the powder velocity. On a qualitative basis, perfume oil is not particularly viscous nor sprayed in a large quantity relative to the mass of powder. Interactions between the powder granules also help to limit the effects of agglomeration (Litster et al, 2004) though equation **(2.17)** may be used to indicate whether the powder velocity would need to be unfeasibly high to ensure a drop controlled regime. If the granules are over saturated, this can be remediated in the design phase by adding additional nozzles at different points to reduce the flow rate (Salman et al, 2007). For a single nozzle flow rate, the powder velocity is:

$$v_p = \frac{3\dot{V}}{2\psi_a d_{droplet} W} = \frac{3 \cdot \left(3.8x10^{-4} m^3 \min^{-1}\right)\left(\frac{1\min}{60s}\right)}{2 \cdot 0.07 \cdot \left(3x10^{-4} m\right)\left(0.615m\right)} = 0.74ms^{-1}$$

The powder velocity in a mixer is typically measured by high speed cameras or positron emission tracking (PEPT). This would be performed at a lab/pilot scale and used with the spray flux to scale up the mixer (Litster et al, 2004). In this theoretical design, the linear velocity of the drum is used as a comparative value to estimate the velocity of the powder. A general guideline states that the powder velocity is typically one magnitude less than the drum linear velocity (Litster et al, 2004). However this provides only a rough estimate and cannot be used to accurately characterise the powder conditions within the drum. In lieu of PEPT, or other monitoring techniques, the design basis of the drum can be made flexible enough to accommodate this uncertainty. The linear velocity of the drum can be found from equation **(2.18).** Using the design rotational speed of the drum, the linear velocity can then be found as 1.07ms^{-1}.

$$v_L = \omega R \tag{2.18}$$

$$v_L = \omega R = \left(2.74 rads^{-1}\right)\left(0.39m\right) = 1.07ms^{-1}$$

The result indicates a similar magnitude between the powder velocity within the drum and the linear velocity for the given spray flux. In order to design flexibility into the drum, the nozzle positions are sited as follows; the first nozzle is sited at a lateral position of 1.31m inside the drum (based on the drum length of 4.8m) and the second nozzle is sited at a lateral position of 2.12m inside the drum. There is a 1m clearance between each nozzle spray zone, and a 0.8m zone on at the outlet and inlet of the drum to prevent product build up. These positions are shown in Figure 5.5.

2.4 Motor Power Requirement for Drum Mixer

The power requirement for the drum mixer is comprised of the energy necessary to rotate the drum at a steady angular velocity, the movement of the powders within the drum and any inherent frictional losses from the driving system (the drive assembly). The power is given by equation **(2.19)** (Masuda et al, 2006).

$$N_m = 2\pi N_s T \tag{2.19}$$

Where: N_m = power requirement to satisfy sum of forces acting on the drum (W), T = total torque requirement for the drum (Nm)

For a horizontal cylinder mixer, the forces acting on the drum comprise the gravitational force of the material at the centre of its mass (the powder bed), the force necessary to rotate the drum on its axis and the frictional force that is a function of the vessel wall and the powder held against the wall by centrifugal force. An empirical relationship can be used to determine the sum of torque requirements for a horizontal rotary cylinder in equation **(2.20)**. This equation relates the Newton Number (on the left hand side) to the Froude Number of the drum (Masuda et al, 2006).

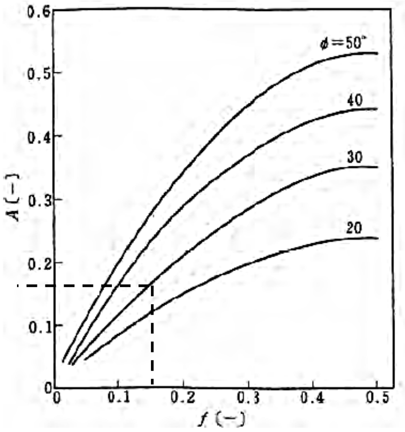
$$\frac{T}{R^3 L_{drum} \rho_b g} = A + B\frac{N_s R}{g} \tag{2.20}$$

Where: T = total torque requirement (Nm), A = dimensionless coefficient that is a function of vessel wall friction coefficient, angle of repose and filling degree, B = dimensionless coefficient that is a function of vessel wall friction coefficient, angle of repose and filling degree

The coefficients A and B are determined from Figures 2.10 and 2.11 as a function of the charge ratio (or volumetric fill ratio) of the drum and the powder angle of repose (for coefficient A) and the wall friction angle (for coefficient B). The actual power requirement for the drum can be determined from equation **(2.21)**, with an assumed motor efficiency of 0.7 (Masuda et al, 2006).

$$N_p = \frac{N_m}{\eta} \tag{2.21}$$

Where: N_p = actual motor specified power (W), η = motor efficiency (0.7)



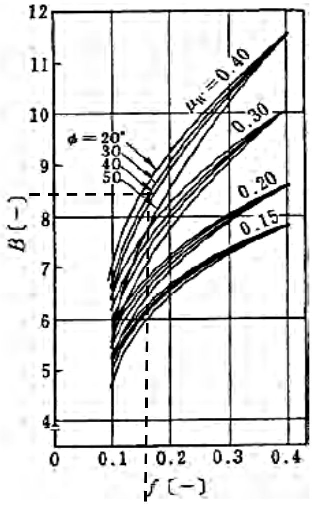


***Figure 2.10**. Coefficient A as function of the filling degree, for different angle of repose (Masuda et al, 2006)*

***Figure 2.11**. Coefficient B as function of the filling degree, for different angle of repose (Masuda et al, 2006)*

The angle of repose is given by Bayly et al, 2017 as 33°, which corresponds to a coefficient A value of 0.18. Similarly, wall friction angle is taken as 22.8° from equation **(2.8)**, which corresponds to 8.3 for coefficient B. Using these values, the power requirement is given by:

$$\frac{T}{R^3 L_{drum}\rho_b g} = A + B\frac{N_s R}{g} \Rightarrow \frac{T}{0.4^3 m^3 \cdot 4.8m \cdot 850kgm^{-3} \cdot 9.81ms^{-2}} = 0.18 + 8.3\frac{\left(0.432revs^{-1}\right)\left(0.4m\right)}{9.81ms^{-2}}$$

$$T = 622.65Nm \approx 623Nm$$

$$N_m = 2\pi N_s T = 2\pi\left(0.432revs^{-1}\right)\left(580Nm\right) = 1688.9W$$

$$N_p = \frac{N_m}{\eta} = \frac{1688.9W}{0.7} = 2412.7W \approx 2.5kW$$

This value may be compared against the supplied motor power of the 36 x 9 SS Munson Machinery rotary mixer provided in Appendix A, which is given as 3HP (2.23kW). The proportionality relationship between the drum volume and power requirement is given by equation **(2.22)** (Perry et al, 2007).

$$N_p \propto L_{drum} D^2 \qquad \textbf{(2.20)}$$

Based on drum volumes, the expected power requirements would be:

Current Design: $P \propto \left(4.8m\right)\left(0.8m\right)^2 = 3.07kW$

Munson Model 36 x 9 SS: $P \propto \left(2.74m\right)\left(0.91m\right)^2 = 2.27kW$

Using this rough scaling relationship, without any powder property data, the trend in the power requirement for the mixer in this design is in line with the expected power requirement as the drum volume increases.

3 AUXILIARY ITEMS

3.1 Auxiliary Item - Pump Selection and Pipeline Design

The mode of operation for pumps may be classified into either centrifugal or positive displacement pumps. Each given pump type has characteristic properties such as response to viscosity, efficiency dependence on pressure and shearing characteristics. In this design, centrifugal pumps are preferred due to the relatively low viscosity of the liquid pumped (perfume oil) and the flexibility in the event of formulation changes. They also tend to be cheaper, easier to maintain due to fewer moving parts and accommodate a smaller area within the plant (Sinnott, 2005).

3.1.1 Pump Design

A schematic drawing of the proposed pipeline for the perfume oil is shown in Figure 3.1. In this design, a detailed plant layout is not available for thoroughfares, exit points and other areas that the pipeline may have to navigate. It is, however, known that the majority of the powder systems in the plant will be gravity fed. The post addition mixer is the final stage in the process, so it is reasonable to assume that this stage will be situated at the ground level of the plant.

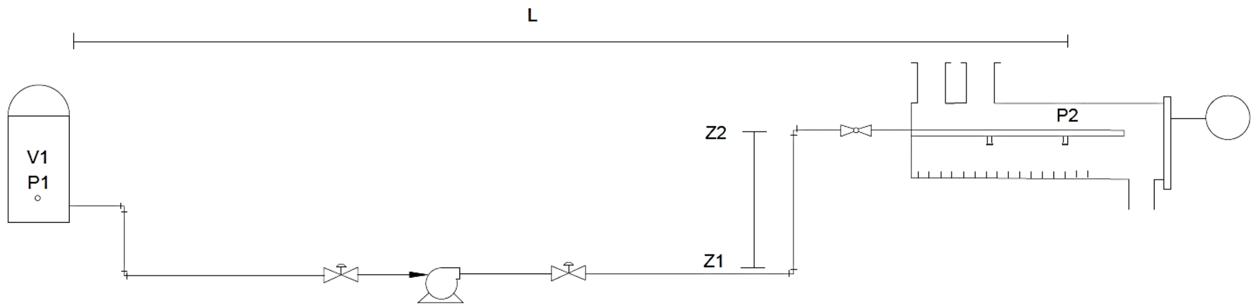


Figure 3.1. *Schematic drawing of pipeline and equipment for liquid addition*

The total pumping head requirement for the pipeline is given by equation **(3.1)** (Sinnott, 2005). In the equation, point 1 corresponds to the level of the storage tank shown in Figure 3.1, while Point 2 is the position of the second spray nozzle sited within the drum mixer (3.21m). The pressure at Point 2, P_2, corresponds to the pressure requirement of the spray nozzles determined previously as 593kPa.

$$h_{pump} = \frac{P_2 - P_1}{\rho g} + \alpha \frac{V_2{}^2 - V_1{}^2}{2g} + (z_2 - z_1) + h_f + \sum h_m + h_{eq} \tag{3.1}$$

In order to determine the pumping head, a series of simplifications can now be made regarding this piping system. Firstly, the pressure head at P_1 is the pressure inside the storage tank. This is taken as the same as atmospheric pressure. Similarly, the velocity at Point 1, V_1, is taken as negligible as the liquid in the tank is stagnant or much less than the velocity at V_2. The head loss due to process equipment (h_{eq}) is taken as 0m, as there is no process equipment in the pipeline. Finally, it is assumed that the storage tank is situated inside the plant itself as it will likely will not take up a large area footprint given the low rate of liquid addition. Based on this, the only elevation loss of the piping system is the distance required to raise the liquid to the sprayer bar position within the mixer (0.4m). In order to account for potential bunding or other framework around the tank or mixer, an elevation head of 1 m was selected. From these axioms, equation **(3.1)** now becomes:

$$h_{pump} = \frac{593x10^3 Pa - 101.325x10^3 Pa}{\rho g} + \alpha \frac{V_2{}^2}{2g} + 1m + h_f + \sum h_m$$

In order to determine the kinetic head of the pipeline, the optimal pipe diameter must be known. The economic pipe diameter for laminar flow in stainless steel pipes (D_i < 1 in) can be found from equation **(3.2)** (Peters, 1991). The total volumetric flow rate and viscosity of perfume oil is given as 1.27x10^{-5}m^{3}s^{-1} and 0.015Pas, respectively (Table 2.4). The optimal diameter is determined as 7.2mm but for simplicity in purchasing, the optimal pipeline diameter is taken as 7.5mm.

$$D_{i,opt} = 3.6\left(\frac{\dot{V}}{0.0283}\right)^{0.4}\left(\frac{\mu}{0.001}\right)^{0.2} \tag{3.2}$$

Where: $\dot{V}$ = flow rate of perfume oil (m^{3}s^{-1}), μ = viscosity of perfume oil (Pas), $D_{i,opt}$ = economic pipe diameter (m)

$$D_{i,opt} = 3.6\left(1.27x10^{-5}\right)^{0.4}\left(0.015Pas\right)^{0.2} = 7.2mm$$

3.1.2 Estimation of Pipe Wall Thickness

The pipe wall thickness may now be determined from equation **(3.3)** (Sinnott, 2005). In this design, the pipeline stresses are low and the perfume oil is not corrosive or abrasive so a thin walled pipe may be used (0.63kgm^{-1}). For stainless steel, the nominal bore of the pipe is selected as 9.53mm which is close to the optimal economic pipe diameter of 7.5mm and allows the pipe to be oversized to compensate for pressure loss in the nozzle. The corresponding outside pipe diameter is 17.2mm. The design stress may be assumed as 100kPa for #312 stainless steel pipe operating at a temperature of 298K (Engineering Toolbox, 2017)(Sinnott, 2005). This gives a pipe wall thickness of 0.27mm.

$$t = \frac{100PD_o}{\left(2x10^4\right)\sigma_d + 100P} \tag{3.3}$$

Where: D_o = outside pipe diameter (m), P = internal pressure of the pipe (kPa), t = wall thickness (m), σ_d = design stress at working temperature (kPa)

$$t = \frac{100(593kPa)(9.53x10^{-3}m)}{(2x10^4)(100kPa) + 100(593kPa)} = 0.27mm$$

This pipe thickness is smaller than the smallest available option in the ASME standard, therefore the smallest available wall thickness will be taken at 1.66mm (Engineering Toolbox, 2017). The corresponding pipeline inner diameter is now taken as 9.53mm. Using the continuity equation **(3.4)** (Sinnott, 2005), and assuming incompressible flow, the pipeline velocity can be calculated as 0.19ms^{-1}.

$$\frac{\dot{V}}{\left(\frac{\pi}{4}\right)\left(D_{i,opt}\right)} = v \qquad \textbf{(3.4)}$$

$$v = \frac{1.27x10^{-5}\,m^3s^{-1}}{\left(\frac{\pi}{4}\right)\left(9.53x10^{-3}\,m\right)^2} = 0.186ms^{-1} \approx 0.19ms^{-1}$$

The type of flow can be determined from the Reynolds Number given by equation **(3.5)** (Sinnott, 2005). The flow regime is laminar, with a Re = 109.

$$\mathrm{Re} = \frac{\rho \cdot v \cdot D_i}{\mu} \qquad \textbf{(3.5)}$$

Where: μ_{po} = viscosity of perfume oil (Pas), D_i = pipe inner diameter (m)

$$\mathrm{Re} = \frac{\rho \cdot v \cdot D_i}{\mu} = \frac{\left(940kgm^{-3}\right)\left(0.19ms^{-1}\right)\left(9.53x10^{-3}\,m\right)}{\left(0.015Pas\right)} = 109 \approx 110$$

Equation **(3.1)** may now be simplified to the following, with the kinetic energy correction factor (α) taken as 2 for laminar flow (Sinnott, 2005):

$$h_{pump} = \frac{593x10^3\,Pa - 101.325x10^3\,Pa}{(940kgm^{-3})\left(9.81ms^{-2}\right)} + 2\frac{\left(0.19ms^{-1}{}_2\right)^2}{2(9.81ms^{-2})} + 1m + h_f + \sum h_m$$

$$h_{pump} = 43.4m + h_f + \sum h_m$$

3.1.3 Head Losses: Friction and Minor Losses

The remaining terms for the pumping head are the head losses due to friction and the minor losses within the pipeline. These are given by equations **(3.6)** and **(3.7)**, respectively (Sinnott, 2005).

$$h_f = 8\phi_f\left(\frac{L}{D}\right)\frac{V^2}{2g} \qquad \textbf{(3.6)}$$

Where: h_f = friction head losses, Φ_f = friction factor, L_p = length of pipe, D_i = inner diameter of pipe

$$h_m = 8\phi_f \left(\frac{L_e}{D_i}\right)\frac{V^2}{2g} \qquad \textbf{(3.7)}$$

Where: L_e = equivalent length of pipe which would produce the same head loss as the equipment piece

The friction factor, *Φf*, can be found from Figure 3.2 as 0.07 for a Re = 110. The actual pipeline length will be determined by a detailed plant layout. Without this, the length from the storage tank positioned within the plant to the outside wall of the drum mixer is assumed as 10m. The distance to the second nozzle within the drum is 3.21m, making the total length of the pipeline 13.21m (Figure 5.5).

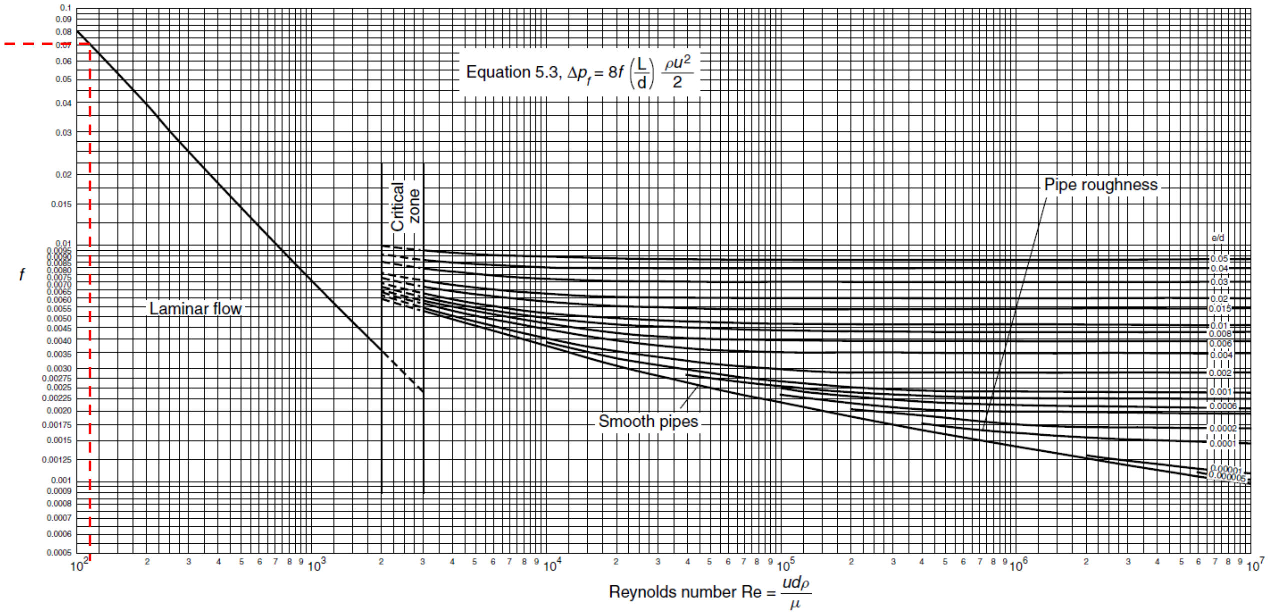


Figure 3.2. *Pipe Friction as a function of Reynolds Number (Sinnott, 2005)*

The frictional head loss for the pipe can then be calculated:

$$h_f = 8\phi_f\left(\frac{L}{D_i}\right)\frac{V^2}{2g} = 8\cdot(0.07)\left(\frac{13.21m}{9.53x10^{-3}m}\right)\left(\frac{(0.19ms^{-1})^2}{2\cdot 9.81ms^{-2}}\right) = 1.428m \approx 1.43m$$

The minor losses in the pipeline includes all pipe fittings, bends and valves. The mixer is fitted with interchangeable sprayer bars so that the liquid addition can be swapped from one to the other in the event of nozzle blockage. The flow may be directed by a ball valve situated at a three point position on the pipeline (Figure 3.3). The pipeline itself uses 5 right angle elbows for a single spraying line (Figure 3.3), as the fluid does not contain significant solids content. Additionally, there are 2 flow control valves and 1 flow direction valve prior to entering the mixer. The equivalent length loss of the nozzles are also considered for the total head loss. These equivalent lengths are summarised in Table 3.1.

__Table 3.1__. Summary of equivalent pipe diameters and fittings for pipeline

Item	Number of Pipe Diameters	Number of Fittings	References
90º elbows (standard radius)	40	5	Sinnott, 2005
Needle Valve	1000	2	Delft University of Technology, 2015
QVVA – SS650067 Nozzle Head	1900	2	Spraying Systems Co, 2016
Ball Valve	18	1	Delft University of Technology, 2015

The minor losses may now be computed from equation **(3.7)**.

$$h_f = 8(0.07)\frac{(5\cdot 40 + 2\cdot 1000 + 2\cdot 1900 + 18)}{(9.53x10^{-3}m)}\frac{(0.19ms^{-1})^2}{2(9.81ms^{-2})} = 648.7m \approx 650m$$

The total pump head requirement is then calculated:

$$h_{pump} = 43.4m + h_f + \sum h_m = 43.4m + 1.43m + 648.7m = 693.55m \approx 700m$$

The hydraulic power requirement for the pump is a function of the flow rate and total pumping head, given by equation **(3.6)** (Sinnott, 2005). The hydraulic power requirement for the pump is 81.97W, though the total power input to the pump depends on the pump efficiency, given by equation **(3.9)**. Based on an assumed efficiency of 0.7, the total power input is calculated as 120W.

$$N_h = \rho g \dot{Q} h_{pump} \qquad \textbf{(3.8)}$$

Where: N_h = hydraulic power (kW), ρ = density of liquid (kgm^{-3}), g = acceleration due to gravity (ms^{-2}), $\dot{Q}$ = volumetric flow rate (m^3s^{-1})

$$N_p = \frac{N_h}{\eta} \qquad \textbf{(3.9)}$$

$$N_h = \rho g \dot{Q} h_{pump} = (940kgm^{-3})(9.81ms^{-2})(1.27x10^{-5}m^3s^{-1})(700m) = 81.97W$$

$$N_p = \frac{N_h}{\eta} = \frac{81.97W}{0.7} = 117.1W \approx 120W$$

3.1.4 Isometric Sketch

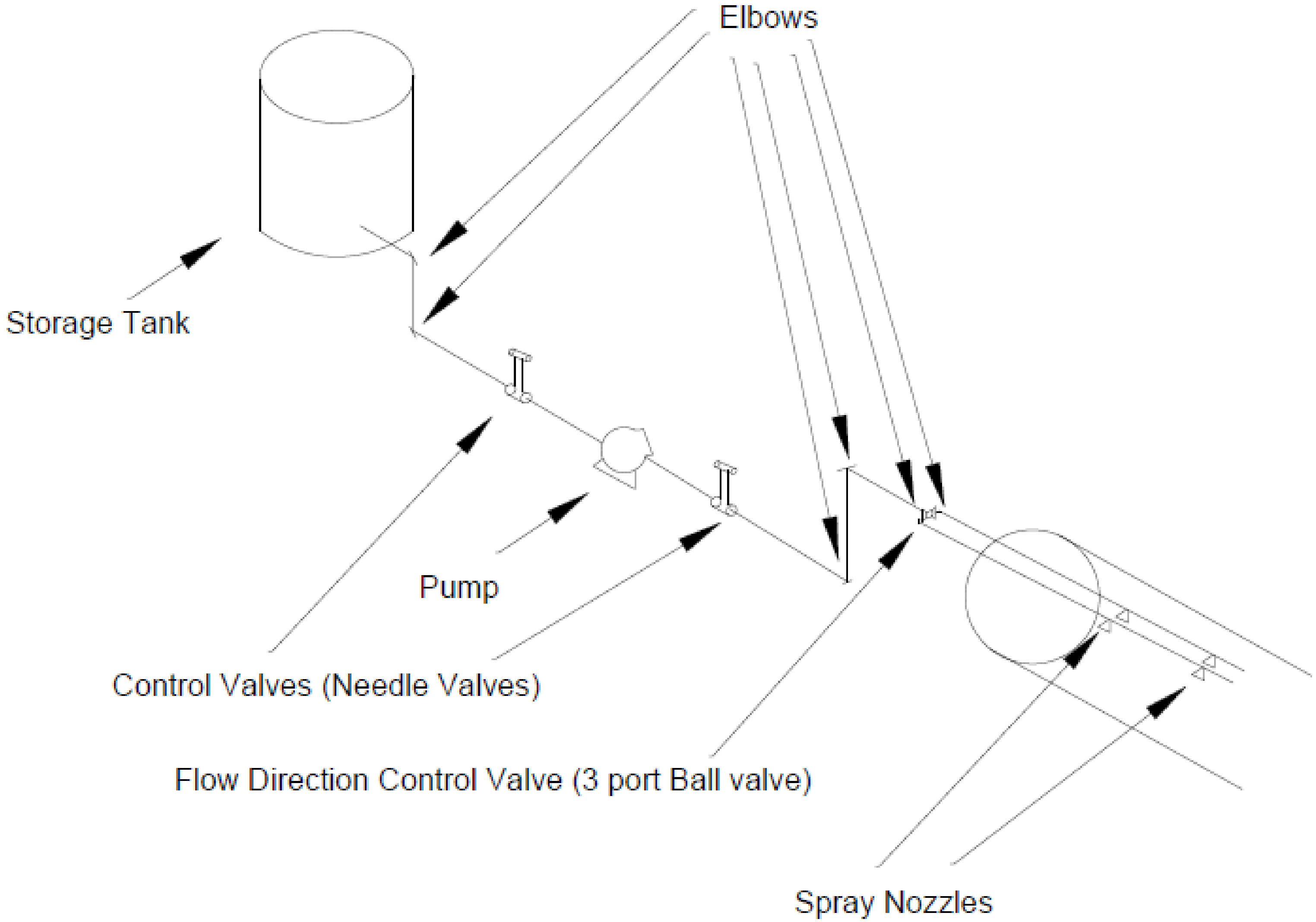


Figure 3.3. *Isometric sketch of piping system for perfume oil liquid addition pipeline*

3.2 Auxiliary Item – Bulk Powder Surge Hopper

3.2.1 Hopper Selection

The surge hopper receives the bulk powder from the fluidised bed dryer and cooler and supplies a consistent feed rate to the drum mixer, remediating any fluctuations. The main selection criteria for hoppers are the materials of construction (e.g. carbon or stainless steel) and the bin geometry, typically conical or wedge hoppers (Maynard, 2013). The feeding of powders through the detergent plant is via gravity systems, with the post addition mixer at the lowest point in the plant layout as the final stage before packaging (Zoller, 2009). A wedge hopper geometry provides a greater cross sectional area and hence storage capacity per unit of height, which is beneficial for limited headroom. However, the current proposal is for a new plant design, and not an existing retrofit. Additionally, flat wall geometry is weaker than cylinder geometry and requires a larger wall thickness and greater supporting structure. Because of this, a conical hopper geometry is selected for the design (Maynard, 2013). The preferred material of construction for the hopper is stainless steel #304-2B, which provides low friction and corrosion resistance from moisture in the detergent powder (Poole, 2017). In addition, the wall friction data used in the experiment is based on a stainless steel lid.

3.2.2 Hopper Design

The following design procedure is based on the output from Brookfield Engineering, 2017, in which a Powder Flow Tester was used to determine the flow function and wall friction for a name brand detergent powder with a bulk density in the range of 700kgm^{-3} – 930kgm^{-3}. The powder characteristics of the proposed Dazzle brand are not available, so an estimation is performed based on this analysis. The considerations for the hopper design are the approximate height of the conical hopper, the desired flow pattern, the hopper angle from the vertical required to achieve this flow pattern and the minimum outlet diameter to prevent arching of the powder (Holdich, 2002). The approximate height of the cylinder is given by equation **(3.10)** (Maynard, 2013).

$$H = \frac{m}{\rho_b \cdot A_c} \tag{3.10}$$

Where: H = cylinder height (m), m = mass to be stored (kg), ρ_b = average bulk density (kgm^{-3}), A_c = cross sectional area of cylinder section (m^2)

The mass to be stored is based on an assumed 2 hour storage time within the hopper. This should be more than sufficient for the plant, as production downtimes longer than 1 hour are associated with maintenance or a serious production fault. In any case, the line will have likely stopped producing the base powder. For an economical design, the cylinder height should be 1-4 times as larger than the hopper diameter (Maynard, 2013). The cylinder height is determined as 5.6m.

$$H = \frac{m}{\rho_b \cdot A_c} = \frac{(6999.24kghr^{-1})(2hr)}{(850kgm^{-3})(2.94m)} = 5.8m$$

The preferred flow pattern for the hopper design is a mass flow system, where the powder enters and exits the vessel on a ‘first in, first out’ basis. This pattern limits segregation of the powder and ensures a homogenous feed to the drum mixer. The condition for mass flow can be found as a function of the wall friction angle and the hopper angle from the vertical (Figure 3.5). In Brookfield Engineering, 2017, the wall friction is plotted as a function of the shear stress for a stainless steel lid (Figure 3.4). The laundry detergent used was a cohesive powder at the lowest consolidation stresses (0.5kPa) with a wall friction angle of 25° (Brookfield Engineering, 2017). This implies flowability issues as the hopper fill reduces. The hopper angle from vertical is therefore designed for this worst case, with an additional safety factor of 3° (Holdich, 2002). The resulting hopper angle from the vertical is 15.5° (Figure 3.5).

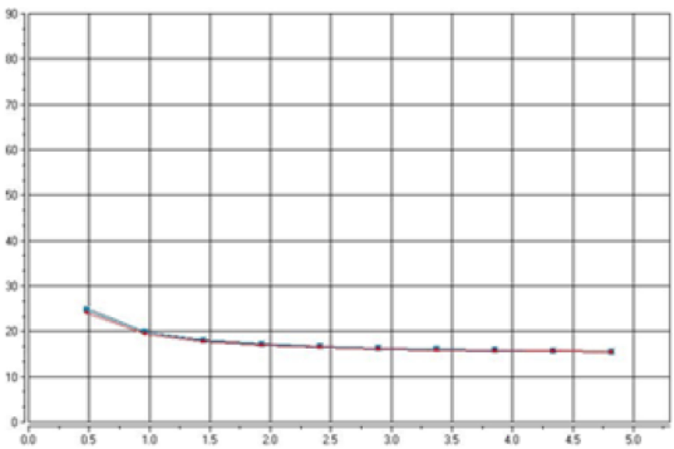


***Figure 3.4**. The wall friction as a function of normal stress. Ordinate: Wall Friction Angle (°), Abscissa: Normal Stress (kPa) (Brookfield Engineering, 2017)*

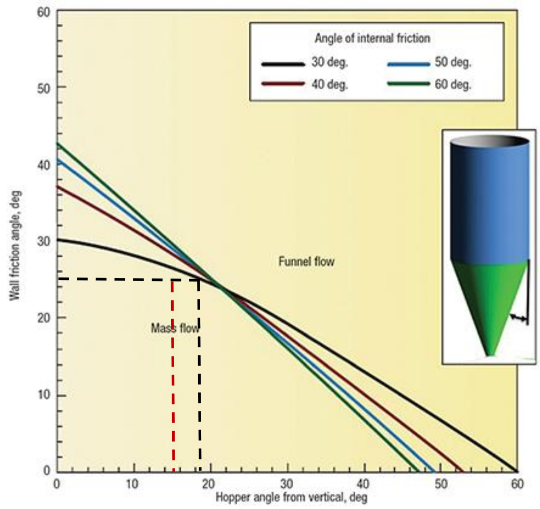


***Figure 3.5**. Theoretical Mass Flow hopper angle from wall friction angle (Maynard, 2013)*

The next consideration for the hopper design is the minimum outlet diameter to prevent the formation of a stable arch at the hopper base. This is given by equation **(3.11)** (Holdich, 2002).

$$D_c = \frac{H(\theta) \cdot \sigma_{crit} \cdot 1000}{\rho_b \cdot g} \qquad \textbf{(3.11)}$$

Where: H(Θ) = is constant value dependent on the hopper geometry, σ_{crit} = critical unconfined yield stress (Nm^{-2}), ρ_b = bulk density (kgm^{-3}), g = acceleration due to gravity ($9.81ms^{-2}$), D_c = minimum outlet diameter to prevent an arch developing (m)

The critical unconfined yield stress can be determined from a flow function curve at the intersection of the arch stress and cohesive strength of the bulk powder. In Figure 3.6, the intersection value is 1.4 which corresponds to a σ_{crit} = 0.32kPa. However, a more conservative value of 0.22kPa is chosen from the lowest possible unconfined yield stress. The value of H (Θ) can be found from Figure 3.7 as function of the hopper angle from the vertical, determined as 2.3. The minimum outlet diameter can then be found:

$$D_c = \frac{H(\theta) \cdot \sigma_{crit} \cdot 1000}{\rho_b \cdot g} = \frac{2.3 \cdot 0.22kPa \cdot \frac{1000Pa}{1kPa}}{850kgm^{-3} \cdot 9.81ms^{-2}} = 0.088m$$

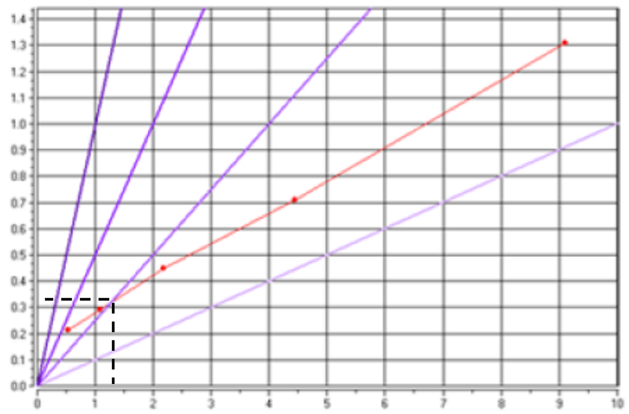


__Figure 3.6.__ A graph of flow function/flowability for the laundry detergent used in Brookfield Engineering, 2017. The red line is the flow function, other lines are reference values. Abscissa: Maximum Principal Stress (kPa), Ordinate: Unconfined yield stress (kPa) (Brookfield Engineering, 2017)

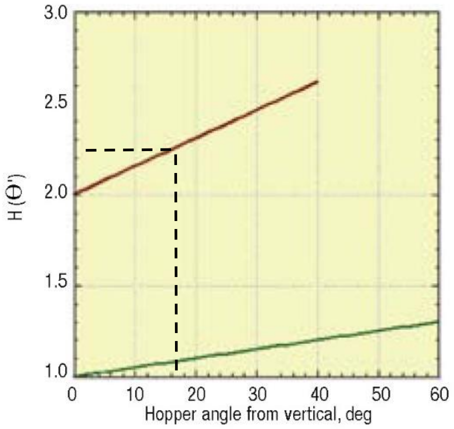


__Figure 3.7.__ A graph of H(Θ) as a function of the hopper angle from the vertical. The red line corresponds to a conical hopper (Maynard, 2013).

Finally, the hopper volume may be approximated by splitting the volume into a closed cone and cylinder (Figure 3.8):

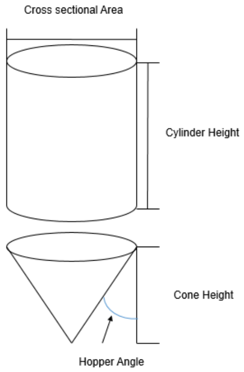


***Figure 3.8**. Schematic diagram of hopper volume calculation*

The cone height can be determined from trigonometry applied to the cone radius, which is found from equation **(3.12)**. If the cone radius is taken as the adjacent line, the height of the cone can be found from **(3.13)**. For an enclosed cone, the volume can be found from **(3.14)**, with the volume of the cylinder applied from equation **(2.3)** where the drum length is substituted by the height of the hopper. The total volume of the hopper can then be determined as 17m^3.

$$R = \left(\frac{A}{\pi}\right)^{0.5} \quad \textbf{(3.12)}$$

$$H_{cone} = R\tan(\theta) \quad \textbf{(3.13)}$$

$$V_{cone} = \pi R^2 \frac{h}{3} \quad \textbf{(3.14)}$$

$$R = \left(\frac{A}{\pi}\right)^{0.5} = \left(\frac{2.94m^2}{\pi}\right)^{0.5} = 0.967m$$

$$H_{cone} = R\tan(\theta) = (0.967m)\tan(15.5) = 0.27m$$

$$V_{cone} = \pi R^2 \frac{h}{3} = \pi(0.967m)^2 \frac{0.27m}{3} = 0.256m^3$$

$$V_{cylinder} = \pi R^2 h = \pi(0.967m)^2(5.8m - 0.27m) = 16.3m^3$$

$$V_{Hopper} = V_{cylinder} + V_{cone} = 0.256m^3 + 16.3m^3 = 16.556m^3 \approx 17m^3$$

3.3 Auxiliary Item – Belt Conveyor

3.3.1 Conveyor Selection

The typical conveyor types used for powder handling are belt, screw or pneumatic conveyors, with bucket elevators often used for vertical transport. For the powder additives, a belt conveyor is preferred as this allows for layering of the additives prior to addition into the drum mixer (Zoller, 2009). Additionally, screw and pneumatic feeders generate fines and produce size reductions of up to 25% via particle breakage. This can adversely affect the median particle size range desired in the end detergent product and presents potential safety hazards (breakage of enzyme polymer coatings, explosion risks) (Salman et al, 2007). The belt should be enclosed to mitigate the formation of dusts within the plant which may cause respiratory or explosive risks.

3.3.2 Belt Design

The following design procedure determines the cross sectional area requirement and belt width for a troughed belt conveyor. The power requirement is not determined in this design as it would require considerable supplementary information about the system and the conveyors are typically bought at a standard rating. The measurable areas are shown in Figure 3.9. The estimation provided assumes that the feed rate of material is uniform and is carried by three equal roll idlers (Fayed et al, 1997).

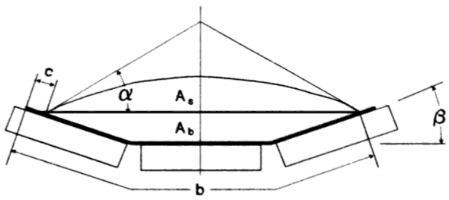


Figure 3.9. *Cross sectional view of a troughed belt and design parameters (Fayed et al, 1997)*

The total cross-sectional area required for the belt can be found from equation **(3.15)**. The belt speed is taken as 0.25ms^{-1} for detergent powders, which in this design is classed as a free flowing material (Woodcock et al, 1988). In industrial practice, the angle of surcharge may be taken as typically 5-15^{o} less than the angle of repose of the powder material which is known as 33^{o} (Bayly et al, 2017). Finally, a normal belt conveyor with no special non-slip devices will have a troughing angle up to 22º (Shamlou, 1988). In this design, a troughing angle of 20º is assumed.

$$\dot{m} = v_{bs} \cdot A_t \cdot \rho_b \tag{3.15}$$

$$A_t = \frac{\dot{m}}{\rho_b \cdot v_{bs}} = \frac{\left(1.94 kgs^{-1}\right)}{\left(850 kgm^{-3}\right)\left(0.25 ms^{-1}\right)} = 9.13x10^{-3} m^2$$

The total cross sectional area is comprised of two components, the trapezoid formed by the idlers (A_b) and the semi-circular portion formed by the angle of surcharge (A_s) (Figure 3.9). These areas are given by equations **(3.16)** and **(3.17)**, respectively. The total area occupied by the powder is given as the sum of these two areas in equation **(3.18)**.

$$A_b = \left[6.45x10^{-4}\,\frac{in^2}{m^2}\right]\left[0.371b + 0.25 + \cos\beta\left(0.2595b - 1.025\right)\right] \cdot \left[\sin\beta\left(0.2595b - 1.025\right)\right] \tag{3.16}$$

$$A_s = \left[6.45x10^{-4}\,\frac{in^2}{m^2}\right]\left[\frac{\pi\alpha_s}{180} - \frac{\sin 2\alpha_s}{2}\right]\left[\frac{0.1855b + 0.125 + \cos\beta\left(0.2595b - 1.025\right)}{\sin\alpha_s}\right]^2 \tag{3.17}$$

$$A_t = A_s + A_b \tag{3.18}$$

An initial value estimate can be made for the belt width, and iterated to determine the desired cross sectional area of the belt. An indicative value of 0.457m (18in) is assumed based on grains and free flowing material (Woodcock et al, 1988).

$$A_b = \left[6.45x10^{-4}\,\frac{in^2}{m^2}\right]\left[0.371(18in) + 0.25 + \cos(20)\left(0.2595(18in) - 1.025\right)\right] \cdot \left[\sin(20)\left(0.2595(18in) - 1.025\right)\right]$$

$$A_b = 8.1x10^{-3} m^2$$

$$A_s = \left[6.45x10^{-4}\,\frac{in^2}{m^2}\right]\left[\frac{\pi 20}{180} - \frac{\sin 2(20)}{2}\right]\left[\frac{0.1855(18in) + 0.125 + \cos(20)\left(0.2595(18in) - 1.025\right)}{\sin(20)}\right]^2 = 1.04x10^{-3} m^2$$

$$A_t = A_s + A_b = 8.1x10^{-3} m^2 + 1.04x10^{-3} m^2 = 9.14x10^{-3} m^2$$

The exact value of belt width is 17.78in (0.452m), though this is unnecessarily precise for specification. As a result, a belt of width 18 inch (0.457m) is selected.

Table 4.1. Design Specification for Primary and Auxiliary Equipment

4 DESIGN SPECIFICATION

DRUM MIXER – PROCESS DATA

Inlet Throughput	Outlet Throughput	Powder Bulk Density	Operating Temperature	Operating Pressure
$8746.24kghr^{-1}$	$8746.24kghr^{-1}$	$850kgm^{-3}$	278K	101.325kPa

DRUM MIXER – EQUIPMENT DATA

Equipment Class	Diameter	Length	Fill	Design Rotational Speed	Tilt Angle	MOC	Residence Time	Minimum Wall Thickness
Horizontal Rotary Drum	0.8m	4.8m	15%	$2.7 rads^{-1}$ (26RPM)	1.5^{o}	#304SS	120 seconds	2.6mm

SPRAY NOZZLE – PROCESS DATA

Fluid	Viscosity	Specific Gravity	Flow Rate
Perfume Oil	0.015Pas	0.94	$3.8x10^{-4}m^{3}min^{-1}$

SPRAY NOZZLE – EQUIPMENT DATA

Model	Number of Nozzles	Spray Pressure	MOC	Spray Angle	Working Distance	Theoretical Spray Coverage
QVVA – SS650067	2	593kPa	#304SS	75^{o}	0.4m	0.615m

MOTOR: DRUM MIXER

Power	Torque	Phase	Voltage	Frequency	Rotational Speed
2.5kW	623Nm	3	230V	50Hz	1800RPM

PUMP

Power	Fluid	Flow Rate	Head	Voltage	Frequency
120W	Perfume Oil	$1.27m^{3}min^{-1}$	700m	230V	50Hz

PIPELINE

MOC	Length	Number of Elbows	Number of Valves	Pipe Wall Thickness	Pipe Outer Diameter
#304SS	10m	5x	2x Needle Valve, 1x Globe Valve	1.66mm	17.2mm

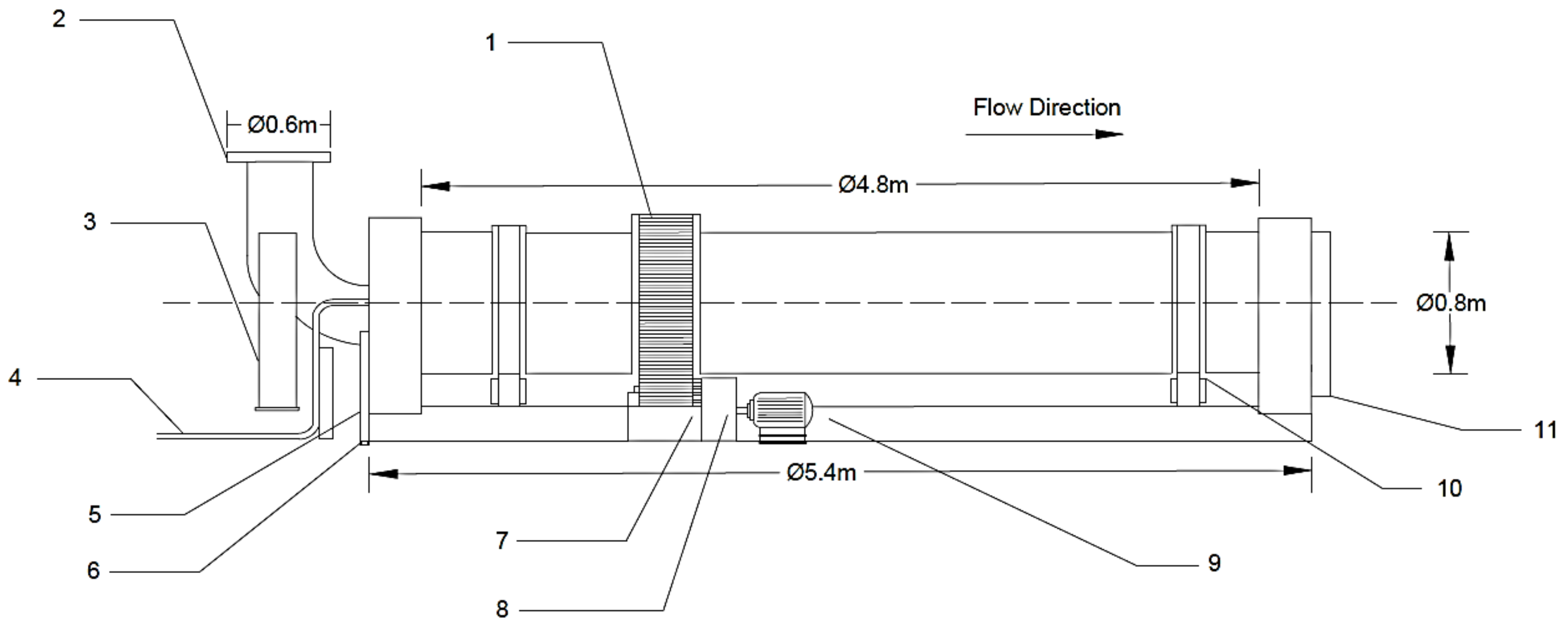
HOPPER

MOC	Diameter	Hopper Height	Hopper Volume	Type	Hopper Half Angle	Minimum Outlet Diameter
#304SS	1.94m	5.8m	17m	Conical	15.5^{o}	0.088m

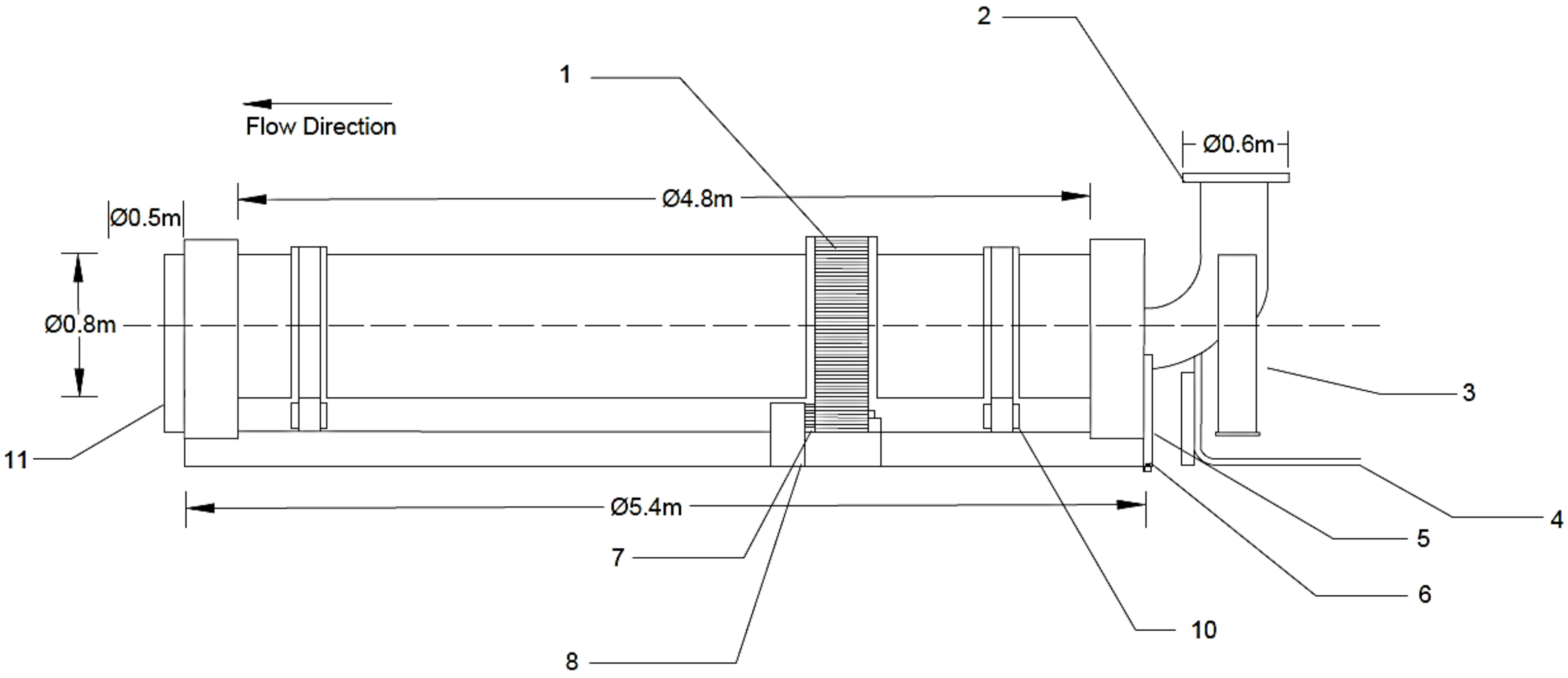
BELT CONVEYOR

MOC	Capacity	Belt Speed	Belt Width	Belt Cross Sectional Area
Natural Rubber	$6999kghr^{-1}$	$0.25ms^{-1}$	0.458m	$9.13x10^{-3}m^{2}$

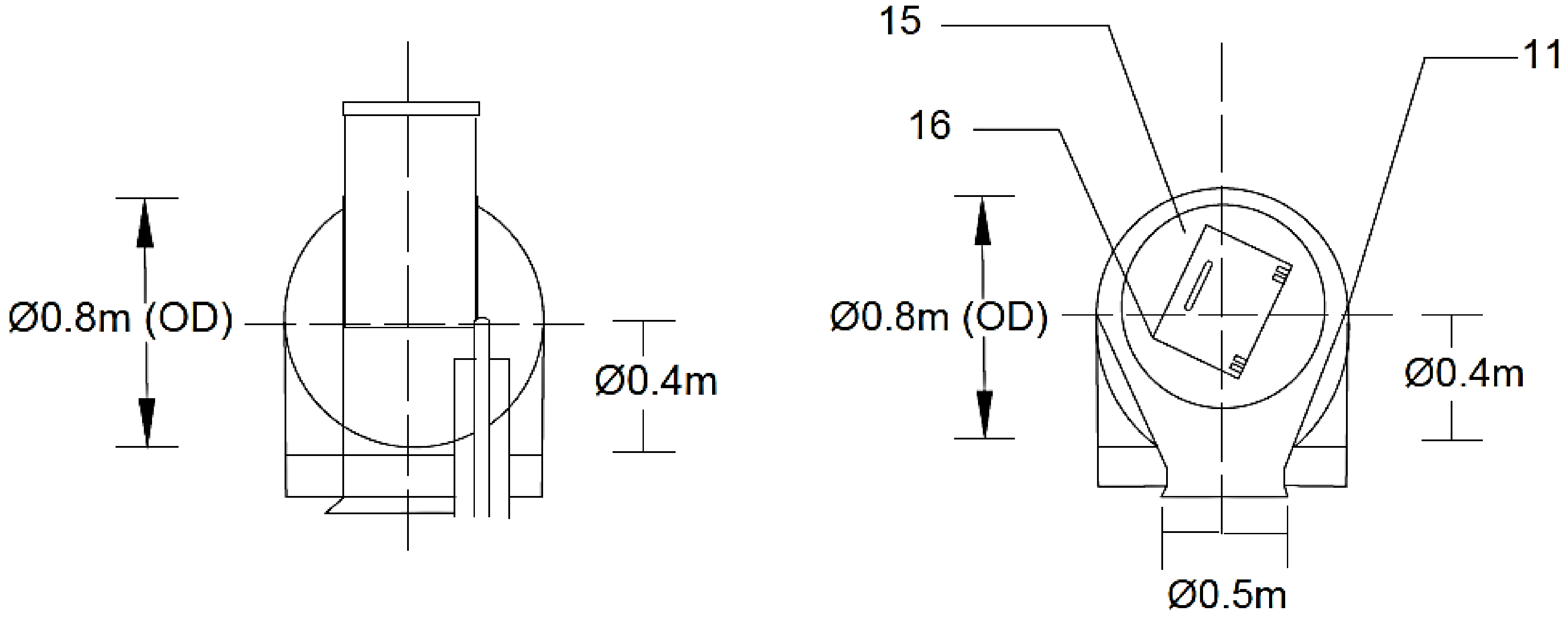
5 ENGINEERING DRAWINGS



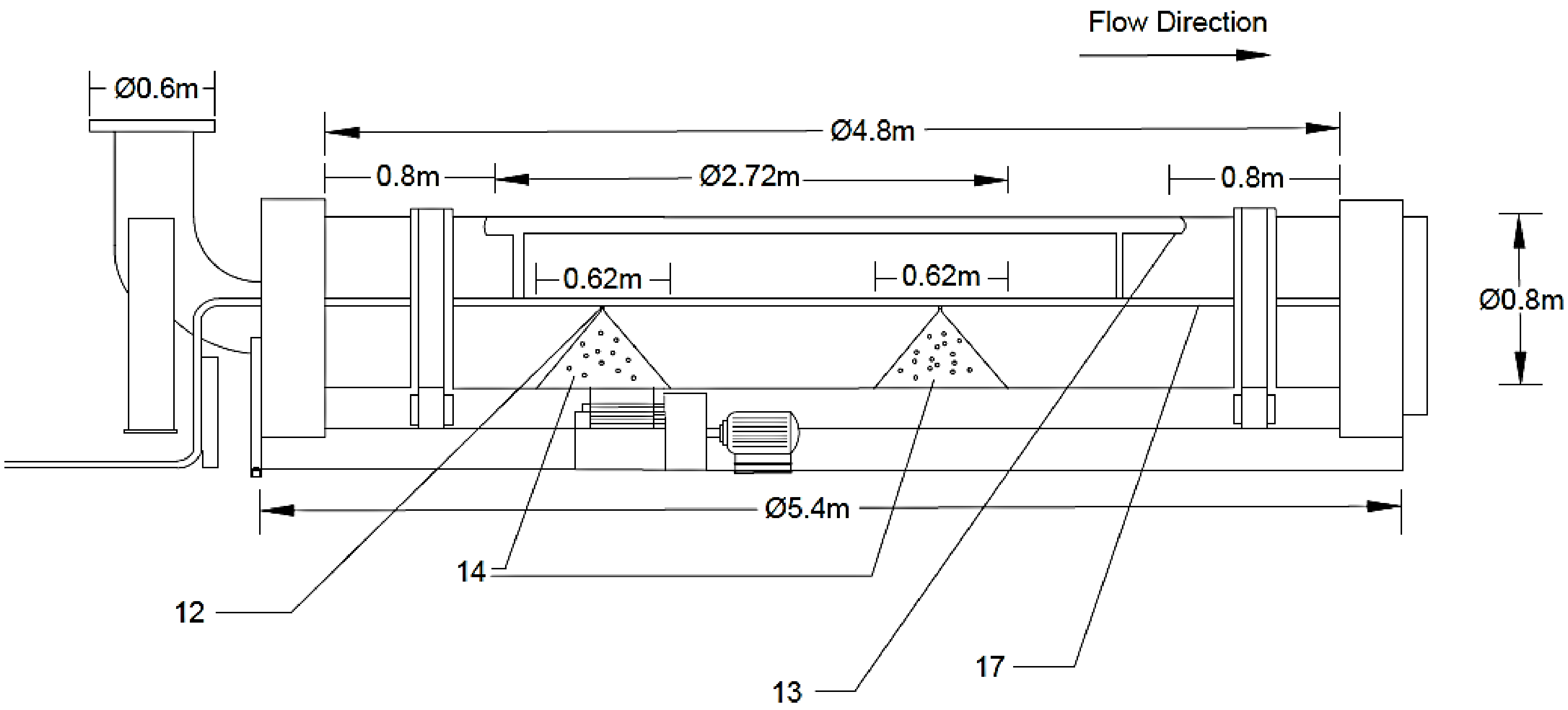
FOLIO 1 OF 5				***Figure 5.1**. Right hand side view of Drum Mixer [External View]*				UNIVERSITY OF LEEDS	
1	Drive Chain	4	Perfume Oil Feed Pipe	7	Hardened Drive Sprocket	10	Rollers	EQUIPMENT NUMBER: DATE:	MI159 19/03/17
2	Inlet Chute	5	Steel Shaft	8	Gear Reducer	11	Outlet Chute	DRAWN BY:	JOSEPH SMITH
3	Inlet Chute Support	6	Grounding Lugs	9	Variable Speed Motor	**SCALE**	Ø1.0m	VERSION:	02



FOLIO 2 OF 5				***Figure 5.2**. Left hand side view of Drum Mixer [External View]*				UNIVERSITY OF LEEDS	
1	Drive Chain	4	Perfume Oil Feed Pipe	7	Hardened Drive Sprocket	10	Rollers	EQUIPMENT NUMBER: DATE:	MI159 19/03/17
2	Inlet Chute	5	Steel Shaft	8	Gear Reducer	11	Outlet Chute	DRAWN BY:	JOSEPH SMITH
3	Inlet Chute Support	6	Grounding Lugs	9	Variable Speed Motor	**SCALE**	Ø1.0m	VERSION:	02



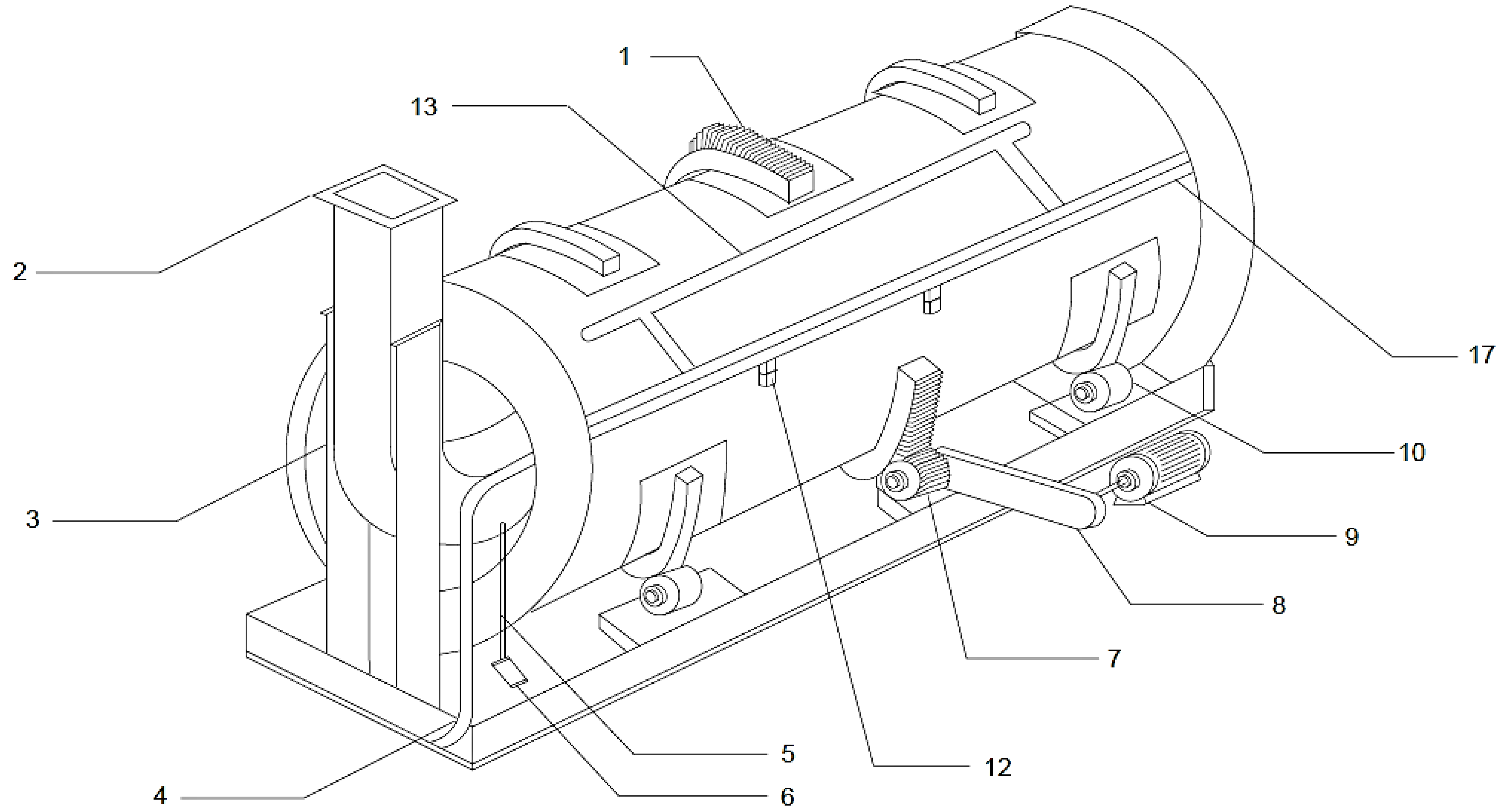
FOLIO 3 OF 5		***Figure 5.3***. *Inlet Drum Wall (Front Face)* ***Figure 5.4***. *Outlet Drum Wall (Back Face)*		UNIVERSITY OF LEEDS	
11	Discharge Chute	**SCALE**	Ø1.0m	EQUIPMENT NUMBER:	MI159
				DATE:	19/03/17
15	Rotary Seal			DRAWN BY:	JOSEPH SMITH
16	Manhole Cover for Maintenance			VERSION:	02



FOLIO 4 OF 5		***Figure 5.5***. *Right hand side view of Drum Mixer [Internal Representation]*		UNIVERSITY OF LEEDS	
12	Spray Nozzle	14	Spraying Zone	EQUIPMENT NUMBER:	MI159
				DATE:	19/03/17
13	Powder Scraper	17	Sprayer Bar	DRAWN BY:	JOSEPH SMITH

ADDITIONAL NOTES: The spray nozzles are sited from the outlet and inlet at a distance equivalent to 15% of the drum length (4.8m). This is to ensure that spray does not impinge on the inlet or outlet and provides sufficient aging time for the powder. A clearance of 1m is given between spray zones to ensure that they do not overlap. The pumping power covers the length up to the second spray nozzle (i.e. Up to half the second spray zone length)

SCALE	Ø1.0m	VERSION:	**02**



FOLIO 5 OF 5				***Figure 5.6**. Isometric view of Drum Mixer [with Internal View]*				UNIVERSITY OF LEEDS	
1	Drive Chain	4	Perfume Oil Feed Pipe	7	Hardened Drive Sprocket	13	Powder Scraper	EQUIPMENT NUMBER:	MI159
				8	Gear Reducer	17	Sprayer Bar	DATE:	19/03/17
2	Inlet Chute	5	Steel Shaft	9	Variable Speed Motor			DRAWN BY:	JOSEPH SMITH
				10	Rollers	**NO SCALE (Illustrative)**			
3	Inlet Chute Support	6	Grounding Lugs	12	Spray Nozzle	NB: Drive and rollers cut away to show internals		VERSION:	02

6 MECHANICAL ENGINEERING DESIGN

The following section provides specification of the materials of construction of the drum mixer and the spray nozzles, in addition to the auxiliary equipment such as the hopper, belt conveyor and pipeline materials. A description is provided of the drive mechanism used to impart torque onto the drum, and any ancillary equipment not considered in the main design (e.g. discharge weirs, trunnions and inlet chute) including necessary safety attachments, maintenance procedures and flexibility in the design for changes in the product formulation. The wall thickness of the vessel is calculated, with the vessel head specified.

6.1 Materials of Construction: Drum Mixer and Spray Nozzle

The drum mixer is fabricated with #304 (18/8) stainless steel as this is most common material used among OEM's and is preferred for hygienic design purposes (Munson Machinery Company Inc, 2016). Other grades available include abrasion resistant steel and #316 stainless steel with molybdenum for corrosion resistance. However, the vessel does not contain chlorides nor is detergent powder particularly abrasive so these types are not selected (Sinnott, 2005) (Paul et al, 2004). The standard materials of construction for spray nozzles include #303 and #316 stainless steel, PVC, PTFE and brass. In industrial applications, the material used should match the rest of the system and attachments. In this case, the nozzle would be fabricated from #304 stainless steel to deliver the liquid additive (BETE Ltd. 2013).

6.2 Drum Mixer: Drive System, Internal and Ancillary Equipment

The preferred motor type is a C-face motor with gear reducer fitted into a chain drive system, with a hardened drive sprocket meshed to the roller chain drive of the drum (Figure 6.1). This chain drive applies torque to the drum which sits on rollers, with an immobile framework providing the supporting structure. The drum ends are fitted with solid weirs that act as retaining dams for the material in the mixer to build up a residence volume. The inlet weir prevents back flow of powder during the mixing process (Pietsch, 2002).

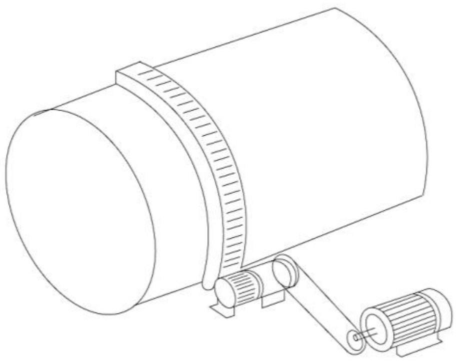


***Figure 6.1**. Proposed drive system*

***Figure 6.2**. Discharging weir for automated cleaning (Munson Machinery Company Inc, 2016)*

Cleaning of the unit can be automated through the use of a discharge weir that lifts remaining material from the drum when the rotation is reversed (Munson Machinery Company Inc, 2016). This reversal of rotation direction is achieved by an adjustable speed drive to swap the motor phase. Additionally, a manhole can be fitted into the drum wall which forms a perfect seal on the inside of the drum but is easily removable for inspection and maintenance (Paul et al, 2004). The drum itself is mounted on trunnions to allow the configuration of the drum tilt angle during operation. These trunnions will gradually wear during operation, requiring tolerance adjustments due to drum settling. Wearing is slow however, due to the low rotational speed. Anti-friction or pillow bearings can be used to mitigate this issue. In the former case, these can be fabricated from solid Nylon or hardened steel (AISI 1045) (Pietsch, 2002) (Munson Machinery Company Inc, 2016). A stationary scraper is attached to remove any excess moist material from the drum walls, and hence limit product build-up (Salman et al, 2007). This scraper should be covered with a wear resistant rubber liner that can be replaced after extended periods of drum operation (Paul et al, 2004). The sprayer bar is supported by appropriate brackets on either end of the drum.

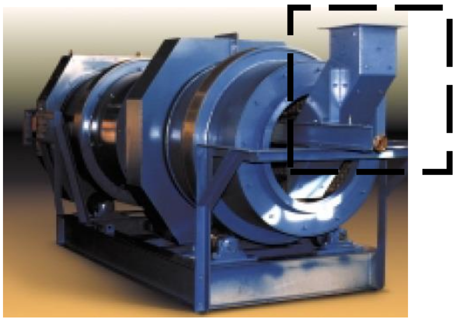


Figure 6.3. *Inlet chute for continuous horizontal rotating drum (Boone, 2015)*

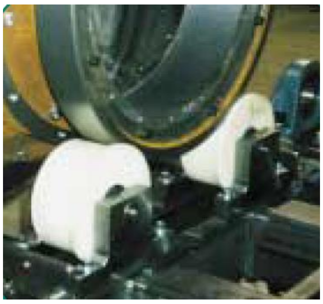


Figure 6.4. *Hardened rollers on which the drum is set (Munson Machinery Company Inc, 2016)*

6.3 Drum Mixer: Dust Mitigation and Explosion Safety, Process Variability

The feeding of the base powder and powder additives is through a completely enclosed inlet chute fitted to the side of the mixer (Figure 6.3) (0.6m diameter), and leaves through an enclosed manifold at the discharge end (0.5m). A rotary seal can provide the necessary mating of surfaces of the inlet chute and drum, and remains stationary as the drum rotates. This seal minimises the potential for dusting from the powder during transport (Munson Machinery Company Inc, 2016) though will gradually wear and require replacement. To minimise electrostatic hazards from agitation of the powder, a steel shaft can be placed that is connected to lay-in grounding lugs at the base of the drum. This shaft will also include carbon brushes to bleed off any electrostatic charge build up (Paul et al, 2004). Due to the atomisation of the liquid within the drum, the spray nozzles should also be suitably earthed from the pipeline. In the event of powder properties changes (e.g. bulk density and angle of repose) due to formulation changes, the residence time of the drum can be altered by pivoting the drum tilt angle with the trunnion. In terms of the rotational speed of the drum, a variable speed motor or inverter may be fitted to alter the speed as necessary (Paul et al, 2004).

6.4 Materials of Construction: Hopper, Belt Conveyor and Pipeline

The hopper design is based on shearing tests performed on a stainless steel lid, for this reason a similar material of construction should be used such as #304-2B stainless steel. This material has the additional benefit of being corrosion resistant and suitable for hygienic design to prevent contamination of the powder. The pipeline used for liquid addition will also be fabricated from #304 stainless steel. An additional sprayer bar is included in the design with a control valve to direct flow to the operational sprayer bar. This can be used as a backup for transport of the liquid in the event of nozzle blockage. For easy removal, quick couplings should be used to allow operators to remove the bar with minimal effort (Boone, 2015). The materials for the conveyor belt should offer anti-static and fire resistance due to the transport of powders and optimally food grade to prevent contamination. Materials available for these requirements include neoprene, natural rubber, styrene butadiene and polybutadiene which are typically used for grain and sugar handling (Shamlou et al, 1988).

6.5 Drum Wall Thickness and Vessel Heads

There are no major internal stresses acting on the drum as in a typical reaction or pressure vessel, as the drum is operated at room temperature and pressure. The primary sources of stress for the vessel are the dead weight loads acting on the vessel. This includes the vessel itself and its contents with external loads such as ancillary equipment and piping attachments. The total weight of a steel vessel, with a cylindrical geometry, is given by equation **(6.1)** (Sinnott, 2005). The direct stress of the vessel is function of the weight of the vessel and its attachments, and the mass of material within the vessel given by the residence time. The total direct stresses are therefore given by equation **(6.2)**. The mean diameter of the vessel, D_m, is given by equation **(6.3)** (Sinnott, 2005). Substitution of **(6.1)** and **(6.3)** into **(6.2)** yields **(6.4)**.

$$W_v = 240 C_v D_m \left(L_{drum} + 0.8 D_m\right) t \tag{6.1}$$

$$\sigma_w = \frac{W_v + W_m}{\pi (D+t) t} = \frac{W_v + (\dot{m}\tau) g}{\pi (D+t) t} \tag{6.2}$$

$$D_m = D + t \, x \, 10^{-3} \tag{6.3}$$

$$\sigma_w = \frac{240 C_v \left(D + tx10^{-3}\right)\left(L_{drum} + 0.8\left(D + tx10^{-3}\right)\right)t + (\dot{m}\tau) g}{\pi (D+t) t} \tag{6.4}$$

The C_v factor can be assumed 1.08 as the attachments for the vessel are relatively minimal and all other values are known. The design stress of #304 stainless steel is taken as $210 Nmm^{-2}$ (BSSA, 2015), which yields a minimum wall thickness of 2.6mm.

$$210 Nmm^{-2} = \frac{240(1.08)\left(0.8m + tx10^{-3}\right)\left(4.8m + 0.8\left(0.8m + tx10^{-3}\right)\right)t + \left(\left(2.43 kgs^{-1}\right)(120s)\right)9.81 ms^{-2}}{\pi((0.8m) + t)t}$$

$$t = 2.59mm \approx 2.6mm$$

The vessel heads are not calculated as the unit operates at atmospheric pressure with the inlet chute fitted to the vessel head (Figure 5.6). In this case, a detailed design would not be representative of the final drum. The vessel head supplied by the manufacturer is an ellipsoidal head, which will be used in this design (Munson Machinery Company, 2016). Finally, the outlet vessel head is connected to the powder discharge, which is open to the atmosphere.

7 EQUIPMENT CONTROL STRATEGY

7.1 Primary Control Objective

The main objective of the control strategy is to ensure that the final product adheres to the specification of a 'white, free flowing powder that is quick to dissolve' and has a bulk density of 850kgm^{-3}. In this, the control scheme should maintain the product quality by managing the liquid and powder rate of addition into the drum mixer. For this reason, feedforward control systems should be employed where possible to remediate disturbances in the system before they affect the product. Given that this is the last stage in the process, any product out of the specification would likely need to be discarded or gradually reintegrated into the mixer in a recycle hopper. The product formulation is achieved by slaving the feed systems of the minor additives to the base powder produced from the fluidised bed cooler. This is the main control loop shown in Figure 7.1. The set point of the materials is based on the formulation specification and material flows provided in Table 1.2. Material is proportioned with multiple loss in weight belt feeders onto an additives weight belt that uses load cells to determine the rate of addition. The additives are layered on a single belt conveyor as shown in Figure 7.1. This simplifies the feeding systems within the plant and maximises space (Zoller, 2009). Similarly, the base powder uses a weight belt with gravimetric measurement of the feed, as volumetric measurements are sensitive to moisture and particle characteristics (Litster et al, 2004). If fluctuations occur in the process, then a variable speed motor changes the conveyor speed on the additives belt. This should be a feed forward control mechanism, as a feedback control would allow disturbances to alter the composition of the product. Additionally, a surge hopper is used to limit the fluctuations in base powder production.

7.2 Controlled and Manipulated Variables

The controlled variables in the finishing stage are the material dosing of base powder, liquid spray on and powder additives into the drum. In terms of the liquid spray ons, these are transported via a pump into the drum mixer and sprayed onto the powder at a fixed rate. Changes in the base powder, as before, result in a corresponding change in the liquid flow rate by altering the needle valve position. This is a feed forward control mechanism that ties the liquid addition to the base powder to the mixer. This can be monitored by a flow meter and transmitter feeding to a remote PLC or computer interface. The valve itself can be pneumatically actuated (Figure 7.1). A feedback control mechanism would again allow the end product composition to be altered before any remediation can take effect. These fluctuations should be anticipated and remediated before the liquid enters the mixer (Litster et al, 2004). The manipulated variables in the feeding systems are summarised in Table 7.1.

***Table 7.1**. Key Process Parameters for Feeding Systems*

Key Process Parameter	Manipulated Variable	Device	Nomenclature
Base Powder Feed	Belt Conveyor speed	Weigh belt feeder/load cells	WI
Powder Additive Feed	Belt Conveyor speed	Weigh belt feeder/load cells	WI
Liquid Additive Feed	Flow Rate, Valve Position	Needle Valve	FI

In addition to the flow and weight measurement, there are two control loops based on the drum mixer performance. In the first, the level of the drum is monitored by a level indicator feeding into a level controller and inclinometer. The inclinometer controls the drum tilt angle, and can be varied to change the residence time of material in the drum. For example, if the liquid rate of addition into the formulation must be higher, then the drum incline can be lowered to increase the time that the powder is exposed to the liquid. The other control loop is based on the rotational speed of the drum. If the formulation specification changes, then the weight indicator from the base powder can alter the rotational speed of the drum to maintain a cataracting regime. The final control loop is monitoring the pumping system and spray nozzles by a pressure indicator. Sudden spikes in pressure can indicate blockages in the pipeline or within the nozzles, forming a feedback control system. This would then be remediated by altering the flow direction to the alternate sprayer bar if the blockage is caused by the nozzles.

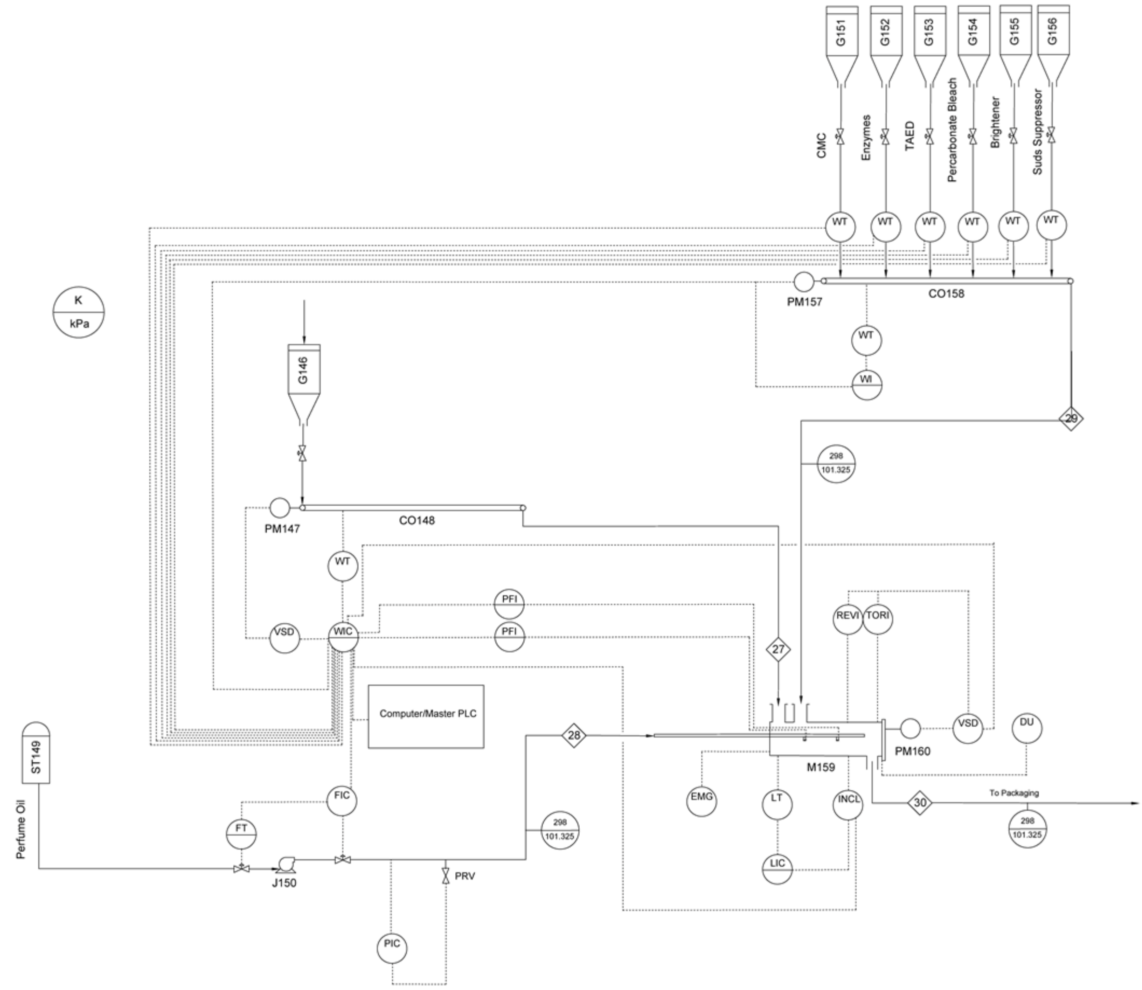
7.3 Instrumentation: Drum Mixer/Spray Nozzle and Ancillary Equipment

The rotary drum is fitted with a torque and tachometer to monitor the applied torque and rotational speed from the motor, which is transmitted to a master PLC or remote computer system. In the event of formulation changes (e.g. bulk density), the rotational speed can be measured and adjusted with the variable speed motor or an attached inverter to maintain a cataracting regime. The residence time within the drum is a function of the drum tilt angle, which can be monitored with an inclinometer and fed to the control system. Component wear, and hence settling of the drum, can be monitored by the inclinometer and adjusted accordingly by altering the incline from the trunnions (Paul et al, 2004). This would also provide early warning for required maintenance on the plant so that this can be scheduled with minimal disruption.

The drum fill level is monitored with a level detection switch for high and low fill volumes – this can be capacitive level measurement device that triggers high and low level alarms when the drum fill volume varies outside of a range (e.g. wider than ±5% of the desired fill level). This could indicate accumulation of product within the drum and potentially other problems such as overspray of the nozzle, improper seals or the formation of layers of powder due to static. Subsequent remediation of the fill level could be achieved by tilting the drum to reduce the residence time or necessary cleaning procedures (Paul et al, 2004).

The spray coverage of the nozzle can be indicated by several disturbances in the system. For direct monitoring, a pressure sensor such as SprayCheck® may be used to indicate nozzle blockages and a loss of coverage of the powder. This may typically be due to accumulation of wetted fines in the nozzle orifice for which the sprayer bar may be repositioned during maintenance if this becomes an issue (Spraying Systems Co. 2013). A non-automatic method of controlling the rate of spray flux is the number and position of nozzles within the drum, particularly if the granule properties vary due to formulation changes (Litster et al, 2004). Finally, a manual kill switch should be securely mounted in an easily accessible position in the event of an emergency (e.g. fire alarm, accident).

7.4 Post Addition Stage – Piping and Instrumentation Diagram



UNIVERSITY OF LEEDS SCHOOL OF CHEMICAL AND PROCESS ENGINEERING PEME3000 DESIGN PROJECT PIPING AND INSTRUMENTATION DIAGRAM		LEGEND									
		C	Control	F	Flow Rate	J	Pump	PM	Motor	TOR	Torque Meter
		CO	Conveyor	G	Hopper	L	Level	PRV	Pressure Relief Valve	T	Transmitter
		DU	Dust Monitor	I	Indicator	M	Mixer	REV	Tachometer	VSD	Variable Speed Drive
VERSION 02 Additional controls added	JOSEPH SMITH	EMG	Emergency Stop	INCL	Inclinometer	P	Pressure	ST	Storage Tank	W	Weight

***Figure 7.1**. Piping and Instrumentation Diagram for Finishing Stage*

7.5 Other Control Systems

Automatic dust monitoring systems should also be present to measure the dust level in the general workspace of the drum mixer. Typical options include light scattering devices that operate at ambient temperature (Omron, 2014). In the event of nozzle blockage, the perfume oil may be diverted from the alternate sprayer bar with a control valve (butterfly) so that production is not affected while the other sprayer bar is cleaned. In the event of plant downtime due to maintenance, the hopper outlet for the powders (base and additives) can be closed with a dome valve.

Table 7.2. *Key Process Parameters for Drum Mixer/Spray Nozzles*

Key Process Parameter	Manipulated Variable	Device	Nomenclature
Residence Time	Drum Tilt Angle, Trunnion Position	Inclinometer	INCL
Rotational Speed	Motor Speed/Inverter Setting	Tachometer, torque meter	REV, TOR
Nozzle Spray Coverage	Nozzle Pressure	Pressure Sensor at nozzle outlet	PI
Fill Volume	Powder Level	Level Switch	LI

7.6 Supplementary Control – Product Quality

In addition to the primary control mechanisms and instrumentation, a selection of sampling procedures may also be used to monitor the product quality of the detergent powder which is required to be “white, free flowing and quick to dissolve”. The powder colour may be analysed with a visual inspection, which can provide insight into any significant production issues (e.g. black marks, significant discolouration) or quantitatively for fine quality control with a spectrophotometer (HunterLab, 2017). This can performed after the finishing stage and prior to packaging either with an in-line measurement tool or with regular sampling by operators on a table top device (HunterLab, 2017). Alternatively, a device may be situated prior to addition of the base powder into the rotary drum. At this stage, discolouration of the powder can be attributed to insufficient neutralization of the LAS acid, overheating of the powder in the fluidised bed dryer or pickup of swarf from the material handling operations. Colour may be also used to indicate insufficient formulation ingredients in the final product, such as the optical brightener. In this case, an additional blending hopper/bin can be added to finishing stage to reblend off-specification product when the process is operating under design parameters. This reduces the potential waste from the process as the powder can be gradually added in a small proportion to the rotary drum (Appel, 1989).

Other properties such as the particle size distribution can be monitored by optical granulometry prior to the rotary mixer and ideally after classification of the powder or alternatively just before the packaging stage. The bulk density of the powder can be measured off-line with an Instron compression testing device. Similarly, the flowability of the powder can be measured quantitatively with a powder flow tester off line by an operator (Appel, 1989) (Brookfield Engineering, 2017).

8 COSTING

NB: This plant is situated in Zdemyslice, Czech Republic and supplies detergent powder to Germany, Austria and Switzerland. Costs are therefore in Euros (€) to simplify cash flow balance sheets in Part 2 report.

The following cost estimation provides the necessary capital costs for the purchasing of the drum mixer and auxiliary items. The capital cost of these units is determined from either directly available costing information or from a scaling of units with similar materials of construction but a different capacity. Other aspects to the capital cost, such instrumentation, the pipeline network and installation costs are determined using the Lang Factor method.

8.1 Drum Mixer Capital Cost

In order to determine the cost of an appropriately sized mixer, the author contacted the manufacturer Munson Machinery Company and was provided with a budgetary proposal (full proposal available in Appendix B). This is based on an inlet volumetric flow rate of $10.3m^3hr^{-1}$ assuming constant bulk density of the material of $850kgm^{-3}$. The model provided, CM-36 x 9-SS Continuous Rotary Mixer, was quoted with a base cost of €75655.50, with optional items bringing the total cost to €83560.50. This proposal was provided in March 2017, so can be taken without need to consider inflation though the €EUR costs are based on a current exchange rate of €0.93:$1 which is subject to fluctuations. A truncated description of the unit and attachments are provided in Table 8.1

Table 8.1. *Budgetary Proposal provided to J Smith. See Appendix B.*

Model	Munson Model CM-36 x 9-SS Continuous Rotary Mixer		
Dimensions	Diameter: 0.91m Length: 2.74m		
Vessel	Fabricated from #304 stainless steel with enclosed inlet chute. Discharge is via adjustable weir opening.		
Throughput	$10.2m^3hr^{-1}$ ($360ft^3hr^{-1}$), based on a bulk density of $850kgm^{-3}$		
Motor and Drive System	3HP (2.2kW), 3 phase, 50Hz, 230V, TEFC, 1800RPM Gear reduced with single strand roller chain and sprocket drive, drive and drum guards		
Support Structure	Structural steel bed frame assembly, carbon brushes and grounding lugs. Drum is supported by alloy steel trunnion ring and roller assemblies		
Total Cost	$81,350.00 (€75655.50)*		
Optional Items	#304 stainless steel automatic clean out weir when rotation is reversed (reversing controller not included)	Cost	$6,000.00 (€5580.00)*
	Hinged inspection door on discharge spout cover (with safety interlock)	Cost	$2,500.00 (€2325.00)*
Total Cost with Optional Items	$89850.00 (€83560.50) **		

*Based on current US-EU exchange rate of 0.93 (10/03/16)
**Does not include CE certification, export, insurance, taxes, installation etc
.***Additional $1,500.00 (€1395.00) brokerage fee required

8.2 Auxiliary Items Capital Costing

The costing for the auxiliary items comprises the powder hopper and belt conveyor as these are the largest units aside from mixer in this design. For a conical, stainless steel hopper, the price was estimated from typical market prices for storage hoppers with a similar volume or tonnage and the same materials of construction. Using a base cost of $2000 for a 10442kg stainless steel conical hopper (Valco Melton. 2016) and applying the six-tenths rule for capacity scaling (York, 2012), the cost was determined as $2300 for a storage capacity of 14000kg.

$$C_B = C_A\left(\frac{S_B}{S_A}\right)^{0.6} \quad \textbf{(8.1)}$$

Where: C_B = Capital cost of design capacity, C_A = Capital cost of reference capacity, S_B = design capacity of unit, S_A = reference capacity of similar unit

$$C_B = \$2000\left(\frac{14000kg}{10442kg}\right)^{0.6} = \$2248 \approx \$2300$$

The belt conveyor cost is a function of the conveyor type, materials of construction, belt width and length. In this design, the duty and length of the conveyor is not available as a detailed plant layout has not been determined. An assumed value of 10m is used, with the calculated belt width at 0.457m (18inches). A base cost for a belt conveyor of this size is given as $14000.00 from 2014 (U.S. Bureau of Mines, 1987) (Matches, 2014). This cost must then be scaled to represent costs in 2016, which may be achieved through the Chemical Engineering Plant Cost Index (Peters, 1991). In 2014, for ‘Process Instruments’ a cost index of 411.9 is used against the 2016 index of 556.8 (Chemical Engineering Magazine, 2017). This yields an estimated current price of $19,000 from equation **(8.2)**.

$$C_{2016} = C_{2014}\cdot\frac{C_{Index2016}}{C_{Index2014}} \quad \textbf{(8.2)}$$

$$C_{2016} = \$14000\cdot\frac{556.8}{411.9} = \$18924.98 \approx \$19000$$

The total purchasing cost of the equipment is therefore:

$$C = \$89850 + \$19000 + \$3200 = \$112050 \approx €104207$$

8.3 Total Costing with Installation and Instrumentation

These costs do not consider CE certification, shipping, insurance, taxes or any other miscellaneous costs associated with the equipment. In addition to these purchased costs, an estimation is provided for installation costs associated with piping, positioning, instrumentation and electrical implementation (York, 2012). This can be estimated using the Lang factors provided in Appendix D, taking a Lang factor of 1.4 for the installation. This brings the total cost to €146,000.00.

$$C_{Total} = €104207\cdot(1.4) = €145889 \approx €146000$$

9 CONCLUSIONS

This work has demonstrated the selection and equipment design procedure for a post addition mixer and the supporting equipment, with auxiliary items included such as a belt conveyor, hopper and pipeline. The mixer selected is a horizontal, continuous rotary drum fabricated from #304 stainless steel with a diameter of 0.8m and a length of 4.8m. The drum rotates at a design rotational speed of 26RPM with a motor power requirement of 2.5kW, where two spray nozzles spray the perfume oil into the drum at a pressure of 593kPa.

The primary limitation in this design was the lack of experimental data to scale the mixer. It is noted in Salman et al. 2007 that purely theoretical designs of drum mixers are not possible. In industry, extensive characterization of the detergent powder properties would be performed - such as the particle rheology, shape, PSD, porosity etc. In order to navigate this, representative values from literature were sourced to complete the design. The drum mixer required a cataracting regime for good liquid distribution, for which the Froude Number was used to scale the mixer to maintain this regime. This approach did not take into account the frictional effect of the vessel wall, or the powder stickiness to determine an appropriate rotational speed. If the design were repeated, pilot scale testing of the powder inside a rotary drum under the design fill volume should be completed to determine the optimal radius and rotational speed.

Characterisation of the powder bed was performed with empirical models provided by York et al, 2003 and powder data provided from Bayly et al, 2017. These models were used in lieu of any experimental data or DEM analysis of the system. The data used to build these models does not match the operating conditions in the current design. Notably, the experiments were performed with glass ballotini with a relatively uniform particle shape and size distribution, which would not be the case for detergent powder. Similarly, the system was performed on a batch basis whereas the current design is continuous and most importantly, did not consider the effect of liquid addition on the powder bed.

Estimation of the drum tilt angle was provided from the Saeman model of a powder bed. The chief limitation in applying this model is the assumption that the powder bed is uniform, which is not the case for a cataracting regime. On top of this, the powder property data used to calculate the inclination angle is derived from the work of Bayly et al, 2017 and may not be representative of the detergent powder produced from the current process. This also became a limitation in characterising the wetting performance of the powder, as the spray flux could only be used without the drop penetration time which would be determined from testing (e.g. Washburn test). Unfortunately, the powder velocity required to achieve the desired spray flux could only be compared against the linear velocity whereas PEPT or high speed camera would be used. In spite of these issues, the final design produced a reasonably sized mixer for the mass flow of material and required a motor power that scaled appropriately with those sourced form rotary drum manufacturers.

The hopper and conveyor belt designs had a similar limitations of requiring practical data regarding the system. In the case of the hopper, this design was only possible through the use of testing results provided from Brookfield Engineering, 2017. The pipeline design was developed without an initial plant layout, so included only the basic design information necessary to deliver the liquid additive to the spray nozzles. If repeated, the pump would be sized according to the layout of a planned pipeline.

10 REFERENCES

Anthony, S. J, Hoyle, W. and Ding, Y. 2004. *Granular Materials: Fundamentals and Applications*. Royal Society of Chemistry: Cambridge UK. pp320-325

Appel, P W. et al. 1989. *US Patent 5133924: Process for preparing a high bulk density granular detergent composition*. Lever Brothers Company: United States. p3

AspenONE, 2016. *5.6.2: Preliminary Drum Design Calculations*. Aspen Process Modelling. pp1-3

Bayly, A, Ghadiri, M, Pasha, M and Behjani, M. A. 2017. Numerical Analysis of segregation of formulated powder mixtures by Discrete Element Method. *Powder Technology*: UK. pp3-5. <u>NB: No issue number is provided as this work is unpublished at the time of writing.</u>

BETE Ltd. 2013. *Standard Fan Nozzle*. [Online] [Accessed 06/03/17] Available from: http://www.spray-nozzle.co.uk/resource-links/catalogue-group/catalogue-by-nozzle-design

Betrand, J. et al. 1991. Powder mixing: some practical rules applied to agitated systems. *Powder Technology,* p68.

Boone, J. R. 2015. *Rotary Drum Blender*. [Online] [Accessed 16/03/16] Available from: http://jrboone.com/rotary-drum-mixers

Brookfield Engineering, 2017. *Laundry Detergent: Powder Flow Tester*. [Online] [Accessed 21/02/17]. Available from: http://www.brookfieldengineering.com/education/applications/powder-laundry-detergent.asp

BSSA, 2015. *Comparison of structural design in stainless steel and carbon steel*. [Online] [Accessed 24/03/17]. Available from: http://www.bssa.org.uk/topics.php?article=125

Callaghan Jr., D C. 1996. *Achieving gentle mixing with a horizontal rotary drum mixer*. Powder and Bulk Engineering: US. pp1-4

Chemical Engineering Magazine, 2017. *Current Economic Trends*. [Online] [Accessed 25/03/17] Available from: http://www.chemengonline.com/current-economic-trends

Coulson J. M. and Richardson, J. F. 1999. *Chemical Engineering Vol 1: Fluid Flow, Heat Transfer and Mass Transfer*. Butterworth-Heinemann: Oxford UK. pp314-374

Davies, R. L. et al. 1985. EP0138410: *Process for the manufacture of coloured detergent powder*. Unilever Plc: European Union. pp3

Delft University of Technology, 2015. *Equivalent Length of pipe bends, valves and filters*. [Online] [Accessed 08/03/17]. Available from: http://www.lr.tudelft.nl/en/

DiSalvo, W. A. et al. 1971. *US Patent 3886098: Manufacture of Free Flowing particulate detergent composition containing detergent.* Colgate-Palmolive Company: United States. p2

Engineering Toolbox, 2017. *Stainless Steel Pipes – Dimensions and Weight ANSI/ASME 36.19*. [Online] [Accessed 24/03/17] Available from: http://www.engineeringtoolbox.com/ansi-stainless-steel-pipes-d_247.html

Fayed, M. E. and Skocir, T. S. 1997. *Mechanical Conveyors: Selection and Operation*. Technomic Publishing: Switzerland. pp107-110

Harnby, N et al. 1992. *Mixing in the Process Industry: 2nd Edition*. Butterworth-Heinemann: Oxford, UK. p400

Holdich, R. G. 2002. *Fundamentals of Particle Technology*. Midland Information Technology & Publishing: UK. pp100-112

HunterLab, 2017. *On-Line Spectrophotometers*. [Online] [Accessed 14/03/17] Available from: https://www.hunterlab.com/on-line-spectrophotometers.html

Litster, J and Ennis, B. 2004. *The Science and Engineering of Granulation Processes*. Springer Science and Business Media: US. pp40-194

Litster, J. 2016. *Design and Processing of Particulate Products.* Cambridge University Press: UK. pp259-261

Marshall, J. S. et al. 2014. *Adhesive Particle Flow: A Discrete-Element Approach*. Cambridge University Press: UK. pp320-328

Masuda, H. et al. 2006. *Powder Technology Handbook, 3rd Edition*. CRC Press: US. pp580-733

Matches, 2014. *Belt Conveyor Cost Estimation*. [Online] [Accessed 21/03/17] Available from: www.matche.com

Mayam, 2006. *Rose Natural Fragrance Oil: Material Safety Data Sheet.* Mayam, Italy. p3

Maynard, E. 2013. *Ten Steps to Effective Bin Design*. American Institute of Chemical Engineers: US.

Mellmann J. 2001. The transverse motion of solids in rotating cylinders: Forms of motion and transition behavior. *Powder Technology* **118**. pp251–270

Mort III, P. R. and Sullivan, M. E. 2004. *United States Patent 6794354B1: Continuous Process for making Detergent composition*. Cincinnati, United States: The Procter & Gamble Company. pp2-10

Munson Machinery Company Inc, 2016. *Rotary Continuous Blender.* [Online] [Accessed 05/03/17] Available from: http://www.munsonmachinery.com/products/mixing/RotaryContinuousBlender/index.asp

Omron, 2014. *Air Particle Sensor*. [Online] [Accessed 08/03/17] Available from: http://www.omron-ap.com/products/family/2067/specification.html

Paul, E. L., Atiemo-Obeng, V. A. and Kresta, S. M. 2004. *Handbook of Industrial Mixing*. John Wiley and Sons Inc.: New Jersey, US. pp1247-1333

Paulsworth, C. 2017. *Choosing a mixer for low-shear batch mixing*. Powder and Bulk Engineering. p3

Perry, R. H. and Green, D. W. 2007. *Perry's Chemical Engineers' Handbook*. McGraw-Hill: US. pp21-120-123

Peters, M.S. 1991. *Plant Design and Economics for Chemical Engineers*. McGraw-Hill Publishing: United States. pp365-400

Pietsch, W. B. 2002. *Agglomeration Processes: Phenomena, Technologies, Equipment*. John Wiley and Sons: US. p161

PNR, 2014. *Spray Engineering Handbook*. [Online] [Accessed 14/02/17] Available from: http://www.pnr.eu/

Poole, C. 2017. *CAPE3300: Process Engineering Operations*. University of Leeds: UK. pp3-7

Salman, A. D. et al. 2007. Granulation. *Handbook of Powder Technology*, Vol **11**. pp3-1390.

Shamlou, P. A. 1988. *Handling of Bulk Solids: Theory and Practice*. Butterworth and Co: London, UK. p162

Sinnott, R. K. 2005. *Chemical Engineering Vol 6: Chemical Engineering Design*. Butterworth-Heinemann: Oxford UK. pp194-228

Spraying Systems Co. 2013. *PulsaJet Automatic Spray Nozzles*. [Online] [Accessed 08/03/17]. Available from: http://www.spray.com/

Spraying Systems Co. 2016. Industrial Hydraulic Spray Products. [Online] [Accessed 01/03/16] Available from: http://www.spray.com/cat75/hydraulic/files/71.html

Stephane, A. 2015. *Experimental study and modelling of hydrodynamic and heating characteristics of flighted rotary kilns*. University of Toulouse: France. pp42-50

U.S. Bureau of Mines, 1987. *Bureau of Mines Cost Estimating System Handbook.* U.S. Department of the Interior: United States. pp19-47

Valco Melton. 2016. *Hopper Feeder Bins*. [Online][Accessed 21/03/17] Available from: http://valcomelton.com/solutions-by-product/hot-melt-dispensing-equipment/automatic-adhesive-filling-systems/product-view/hopper-feeder-bins

Weinkotter, R and Gericke, H. 2000. *Mixing of Solids*. Springer Science + Business Media: Switzerland. pp100-150

Woodcock, C. R. and Mason, J. S. 1988. *Bulk Solids Handling*: *An Introduction to the Practice and Technology*. Springer and Science Business Media: US. p375

York, D. W. 2012. *PEME3200 Design Project: Introduction to Project Cost Estimation*. University of Leeds: UK. p6

York, D. W. et al. 2003. Investigation of the Granular Behaviour in a rotating drum operated over a wide range of rotational speed. *Trans IChemE,* Vol ***81***, Part A.

Zoller, U. 2009. *Handbook of Detergents Part F*: *Production*. CRC Press: US. pp324-35

APPENDIX A

Estimation of Drum Tilt Angle – Other Models

Table A.1. *Empirical models used to predict the drum angle (Stephane, 2015)*

Model	Equation	Reference	Equation Number
Sullivan et al.	$\tau = \dfrac{0.03102\left(L8^{0.5}\right)}{DN\tan(\alpha)}$	Masuda, et al. 2006	**(A.1)**
Perry and Chilton	$\tau = \dfrac{0.23L}{DN^{0.9}\tan(\alpha)}$	Mujumdar, 2014	**(A.2)**
Friedman and Marshall	$\tau = \dfrac{0.03344L}{DN^{0.9}\tan(\alpha)}$	Mujumdar, 2014	**(A.3)**
Van Krevelan and Hoftijzer	$\tau = \dfrac{0.13L}{ND\tan(\alpha)}$	Mujumdar, 2014	**(A.4)**

Where: L = drum length, D = drum diameter, N = rotational speed (rpm), α = drum angle of inclination, τ = mean residence time

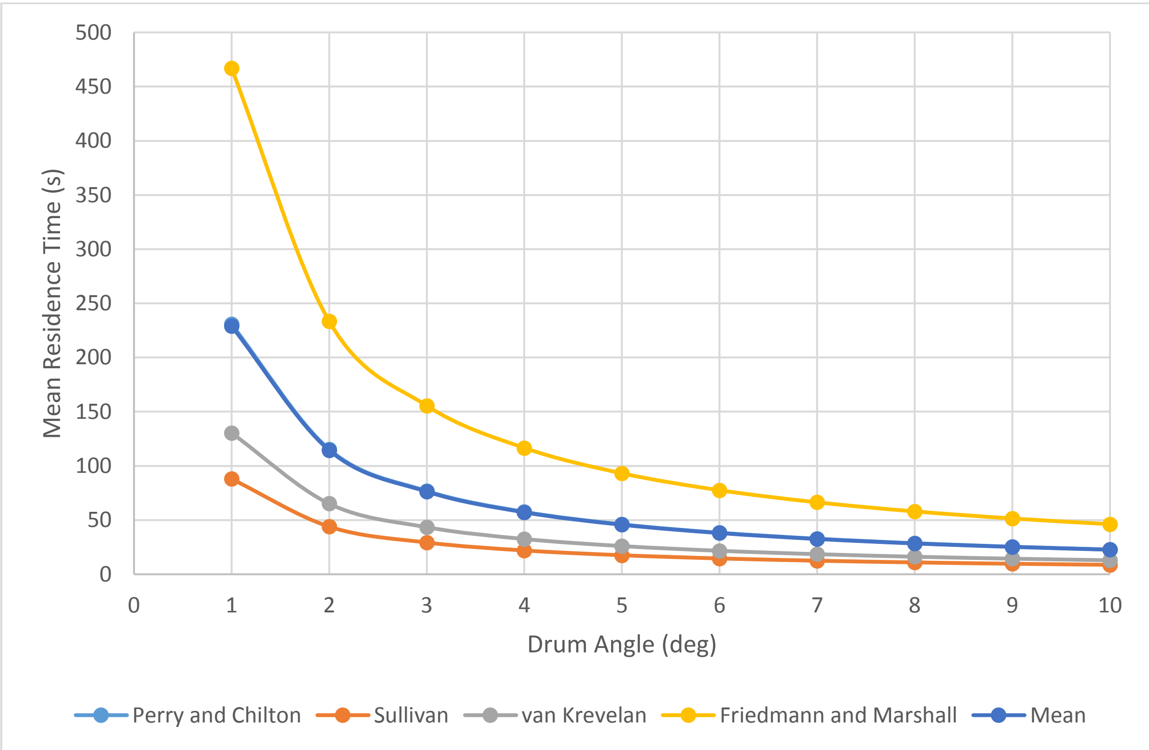


Figure A.1. *The variation in mean residence for the different models. The mean of each model closely overlaps Perry and Chilton.*

APPENDIX B

Munson Continuous Rotary Mixer Enquiry and Budgetary Proposal provided to Joseph Smith

MUNSON

CUSTOMER CHECKLIST FOR CONTINUOUS MIXERS

COMPANY		CONTACT	Joseph Smith			DATE	
STREET		TITLE	Mr				
CITY	Leeds	STATE		ZIP		Country	
PHONE		FAX		EMAIL			

HOW DID YOU HEAR ABOUT MUNSON: Supporting Engineers

PRODUCT CHARACTERISTCS

Material name(s)	(1)	Detergent Powder	(2)		(3)	
Bulk density	(1)	850kgm-3	(2)		(3)	
Percentages	(1)		(2)		(3)	
Particle size	(1)	400-600 micron	(2)		(3)	
Moisture content	(1)	4.50%	(2)		(3)	
Consistency	(1)	Free flowing	(2)		(3)	

Is the material friable?	x	Yes		No	Is the material abrasive?		Yes	x	No
Is the material explosive?		Yes	x	No	Is the material toxic?		Yes	x	No
Is the material heat sensitive?	x	Yes		No	Is the material corrosive?		Yes	x	No
Is the material free flowing?	x	Yes		No	Is the material interlocking?		Yes	x	No
Does the material fluidize?		Yes	x	No	Is the material hygroscopic?	x	Yes		No

Is there a liquid addition?	x	Yes		No	(1)	0.05%$	(2)	0.05$%		wt	vol
Liquid addition name(s)	(1)	Perfume Oil	(2)	Suds Supressor							

MIXER REQUIREMENTS

Mixer/agitator type	x	Rotary (gentle/batch)		Hi-intensity		Double Ribbon		
How is the material fed?	from a belt conveyor ideally into an enclosed chute attachment. Liquids are pumped via s							
Required rate: Cubic feet / hour	10.29		Pounds per hour					
Residence time required	120s							
Desired results (coating, agglomerating, etc	coating, drop controlled nucleation (no agglomeration)							
Heating / cooling jacket required?		Yes	x	No	Specifics			
How are you currently mixing?	N/A							
Electrical requirements?	Voltage	230	Hz	50	NEMA	C	TEFC	
	TEXP (Group Div)				Washdown			
Construction materials		Carbon Steel	x	304 Stainless steel			316 Stainless steel	
		Abrasion resistant		Other	Finish			

Special notes: Liquids are sprayed from spray nozzles on a spray bar

Cleaning should be simple as possible - attachment weirs for automated cleaning

Explosion protection through earthing due to powders, attachment steel shaft

PROPOSAL

NUMBER: Budgetary proposal
DATE: 3/7/2017
PAGE NO. Page 1 of 6
TERMS: 1/3 down payment, net prior to shipping or approved L.O.C.*
ExWorks, Utica NY, U.S.A.

TO: Leeds – United Kingdom
Att: Joseph Smith
West Yorkshire, United Kingdom
C/O: Luis Rosales – REG Process
Authorized MUNSON Representative

DESCRIPTION	PRICE
<u>MUNSON MODEL CM-36 x 9-SS CONTINUOUS ROTARY MIXER</u> <u>SIZE:</u> - Approximately 370 cubic foot per hour blending capacity based on 2 to 2-1/2 minute product residence time based on product bulk density of 53 lbs/ft³ provided by customer (** see note on page 4) <u>BLENDING VESSEL:</u> - Bolted and welded construction, #304 stainless steel product contact surfaces - Vessel dimensions of 36" diameter and 108" length - Highly efficient computer designed progressively pitched mixing flights - Internal mixing flights are bolted to brackets that are continuously welded to the wall drum wall <u>INLET:</u> - Product charge via stationary inlet chute <u>DISCHARGE:</u> - Discharge is via adjustable weir opening with gasketing - #304 stainless steel housing includes removable end panel for access to internal surfaces for cleaning (price for optional hinged door on discharge cover shown on page 3) <u>INLET SEAL:</u> - Dust-free design employs nitrile elastomer mechanical face seal rotating against a precision machined #304 stainless steel surface	

By: Darren E. Woods

Darren E. Woods, General Sales Manager

MUNSON® MACHINERY CO, INC.
210 SEWARD AVE. UTICA, NY 13502
PHONE (315)797-0090OR(800)944-6644– EXT283
FAX (315)797-5582
WEB PAGE: www.munsonmachinery.com
E-MAIL: dwoods@munsonmachinery.com

SEE LAST PAGE FOR TERMS AND CONDITIONS OF SALE

PROPOSAL

NUMBER: Budgetary proposal
DATE: 3/7/2017
PAGE NO. Page 2 of 6
TERMS: 1/3 down payment, net prior to shipping or approved L.O.C.*
ExWorks, Utica NY, U.S.A.

TO: Leeds – United Kingdom
Att: Joseph Smith
West Yorkshire, United Kingdom
C/O: Luis Rosales – REG Process
Authorized MUNSON Representative

DESCRIPTION	PRICE
<u>DISCHARGE SEAL:</u> - Nitrile elastomer mechanical face seal rotating against inside of the #304 stainless steel surface <u>MOTOR:</u> - 3 HP, 3 phase, 50 Hz, 230v, TEFC, 1800 RPM (other voltages available, may impact price) <u>DRIVE:</u> - Gear reducer with single strand roller chain and sprocket drive, drive and drum guards included, carbon steel construction (#304 stainless steel guards available at an additional cost – please advise if required) - Drum rotation is approximately 10-12 RPM <u>SUPPORT STRUCTURE:</u> - Unit supported by structural steel bed frame assembly, carbon brushes and grounding lugs included - Mixing drum supported by alloy steel trunnion ring and roller assemblies <u>FINISH:</u> - 2B (or equivalent) interior surface finish (where available), all welds ground clean with no splatter (higher weld finishes available at an additional cost – please advise if required) - Exterior non-stainless steel surfaces cleaned, primed and painted with Steel-It® paint EQUIPMENT PRICE EACH:	$81,350.00

By:

Darren E. Woods, General Sales Manager

MUNSON®

MUNSON ® MACHINERY CO, INC.
210 SEWARD AVE. UTICA, NY 13502
PHONE (315)797-0090OR(800)944-6644– EXT283
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WEB PAGE: www.munsonmachinery.com
E-MAIL: dwoods@munsonmachinery.com

SEE LAST PAGE FOR TERMS AND CONDITIONS OF SALE

PROPOSAL

NUMBER: Budgetary proposal
DATE: 3/7/2017

TERMS: 1/3 down payment, net prior to shipping or approved L.O.C.*
ExWorks, Utica NY, U.S.A.

TO: Leeds – United Kingdom
Att: Joseph Smith
West Yorkshire, United Kingdom
C/O: Luis Rosales – REG Process
Authorized MUNSON Representative

DESCRIPTION	PRICE
OPTIONS:	
- **MODEL CM-48 x 12-SS** - same as above, except product residence time estimated to be 5-6 minutes, vessel dimensions of 48" diameter and 144" length, 7-1/2 HP motor, adjustable spring pressure radial seal assembly with Teflon® sealing materials (replaces nitrile face seal used on smaller model), equipment price	$111,000.00
- #304 stainless steel automatic clean-out weir, discharges remainder of material when unit is reversed (requires reversing controller, to be supplied by others), add	$6,000.00
- Hinged inspection door on discharge spout cover, includes safety interlock, add	$2,500.00
Please note: this proposal does not contain costs for any products or services not contained herein, including but not limited to the following: - CE Certification - Export crating - Shipping / insurance - Duties / taxes / etc	
Time required for approval drawings: approximately 2-3 weeks	
Shipment: approximately 18-20 weeks after return of final, signed approval drawings and receipt of 1/3 down payment, predicated upon availability of required materials and components. Pricing valid for 30 days. Shipment time subject to change 30 days from quotation date and may vary depending on shop load at time of order. This quotation does not include controls or motor starter unless otherwise stated above.	

By: Darren E. Woods

Darren E. Woods, General Sales Manager

MUNSON® MACHINERY CO, INC.
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PHONE (315)797-0090 OR (800)944-6644– EXT283
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PROPOSAL

NUMBER: Budgetary proposal
DATE: 3/7/2017
PAGE NO. Page 4 of 6
TERMS: 1/3 down payment, net prior to shipping or approved L.O.C.*
ExWorks, Utica NY, U.S.A.

TO: Leeds – United Kingdom
Att: Joseph Smith
West Yorkshire, United Kingdom
C/O: Luis Rosales – REG Process
Authorized MUNSON Representative

DESCRIPTION	PRICE
This quote does not include any costs associated with export from the continental United States; pricing for export can be provided separately. All prices are in US Dollars; Munson cannot be held responsible for fluctuations in international currency exchange rates. International payment terms: 1/3 down payment, balance prior to shipment, <u>or</u> irrevocable, confirmed Letter of Credit lodged at any national bank New York, effective for a period of not less than 180 days, payable at sight with all freight and forwarding arrangements by Munson Machinery. Please note that the information contained in this quotation supersedes any and all previous information, data, and technical specifications. Purchased components (drive, motor, etc.) are covered under the terms of those specific manufacturer's implied warranty conditions and periods. There is a $1,500.00 banking/broker fee for all letters of credit. Any amendments to the original L.O.C. will add an additional $250.00 to this fee. If a bank guarantee is requested, a processing fee of $750.00 *or* 3% of the amount that is guaranteed, whichever is greater, must be charged. These charges represent our actual bank and/or export broker costs for these services and are non-negotiable. Munson cannot be held responsible for variations in materials and/or process, and cannot guarantee applications suitability, power requirements or throughput rates without first testing in our lab unit or evaluating a sample of the material to be processed. **Throughput rate & residence time projections are mathematically calculated and therefore may vary slightly from actual. *Please note the Terms & Conditions shown on the last page of this proposal – any deviation from these Terms must be agreed upon prior to placing an order.*	

By: Darren E. Woods

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APPENDIX C

Table C.1. *General Design Guidelines for Rotary Drum Mixer*

Property	Value		Unit	Reference
Diameter [D]	Typical Range	Typically 1-6	m	AspenONE, 2016
Length [L_{drum}]	Typical Range	Typically 2-10 times [D]	m	Litster et al, 2004
Drum Tilt Angle	Typical Range	0 – 10 from the horizontal	deg	Perry et al, 2007
Residence Times	Typical Range	120s-300s	s	Litster et al, 2004
	Selected Value	120s	s	Davies et al, 1985
Drum Loading	Typical Range	0.1-0.3	dimensionless	Salman et al, 2007
	Selected Value	0.15	dimensionless	Litster et al, 2004
Froude Number	Typical Range	0.3-0.5	dimensionless	Litster, J. 2016
	Selected Value	0.3	dimensionless	Technical Consultancy, Litster, J. 2016
Flow Regime	Cataracting		-	Litster, J. 2016
Rotational Speed	(0.35 – 0.5)N_{cr}		rpm	Salman et al, 2007
	2 – 30rpm ~15rpm		rpm	DiSalvo et al, 1971 Callaghan Jr, 1996
Power Consumption	P α LD^2		-	Perry et al, 2007

APPENDIX D

***Table D.1**. Lang factor installation numbers (Peters, 1991)*

Additional Costs	Instrumentation	0.2
	Piping	0.7
	Positioning	0.4
	Electrical	0.1